Network Science

Concepts, Examples, and Exercises

Zahra Rezaei

2026-03-29

Table of contents

1 Course Overview **5**
1.1 Welcome . . . 5
1.2 Required packages . . . 5
1.3 Current chapters . . . 5

2 Random Graphs **6**
2.1 Introduction . . . 6
2.2 Learning goals . . . 6
2.3 A beginner-friendly motivation . . . 6
2.4 Why the binomial distribution appears . . . 7
2.5 The Erdős–Rényi models . . . 8
2.5.1 The model $G(N, L)$ . . . 8
2.5.2 The model $G(N, p)$ . . . 9
2.5.3 Comparing $G(N, L)$ and $G(N, p)$ . . . 12
2.6 Average degree . . . 12
2.6.1 A quick numerical example . . . 12
2.6.2 Why average degree matters in random graphs . . . 13
2.6.3 Average degree in $G(N, L)$ . . . 14
2.6.4 Average degree in $G(N, p)$ . . . 14
2.6.5 A second way to understand the formula . . . 15
2.6.6 A quick numerical example in $G(N, p)$ . . . 16
2.6.7 Dense and sparse regimes . . . 16
2.6.8 Comparing the two models . . . 16
2.6.9 Summary . . . 17
2.7 Degree distribution . . . 17
2.7.1 A small example . . . 17
2.7.2 From node degrees to degree counts . . . 18
2.7.3 Plotting the degree distribution . . . 19
2.7.4 Degree distribution in the random graph model $G(N, p)$ . . . 20
2.7.5 Interpreting the formula . . . 20
2.7.6 Comparing one realization with the theoretical distribution . . . 20
2.8 Poisson approximation in sparse large graphs . . . 23
2.8.1 Why this approximation appears . . . 23
2.8.2 Class example: binomial distribution and Poisson approximation . . . 23
2.9 Clustering coefficient . . . 24
2.9.1 Local clustering coefficient . . . 24
2.9.2 A fixed node with three different neighborhood structures . . . 25
2.9.3 Example 1 . . . 26
2.9.4 Example 2 . . . 27
2.9.5 Example 3 . . . 27
2.9.6 Summary table for the three examples . . . 27

2.9.7 What clustering is really measuring . . . 28
2.9.8 Average clustering coefficient of a graph . . . 28
2.9.9 Why clustering is important in network science . . . 28
2.9.10 Expected clustering in $G(N, p)$ . . . 28
2.9.11 Why this result matters . . . 29
2.9.12 Numerical check by simulation . . . 29
2.9.13 Summary . . . 31
2.10 Core formulas to remember . . . 31
2.11 Class example: generate and plot a random graph . . . 32
2.12 Interactive explorer . . . 32
2.13 In-class questions . . . 33
2.14 End-of-chapter exercises . . . 33
2.14.1 Part I. Mathematical exercises . . . 33
2.14.2 Part II. Conceptual and analytical exercises . . . 36
2.14.3 Part III. Simulation and coding exercises . . . 37
2.15 Suggested mini-project . . . 39
2.16 Concluding remarks . . . 39

3 The Evolution of Random Graphs 40
3.1 Introduction . . . 40
3.2 Learning goals . . . 40
3.3 Connectedness and components . . . 40
3.3.1 Connected graph . . . 40
3.3.2 Connected component . . . 41
3.3.3 Size of a component . . . 41
3.3.4 Isolated vertex . . . 41
3.3.5 Example: component structure . . . 42
3.3.6 Component-size table . . . 42
3.4 Giant component . . . 43
3.4.1 Definition . . . 43
3.4.2 Why the giant component matters . . . 43
3.5 The evolution of a random graph . . . 43
3.5.1 Evolution through snapshots . . . 44
3.6 Regimes of random-graph evolution . . . 44
3.6.1 Subcritical regime . . . 45
3.6.2 Critical regime . . . 45
3.6.3 Supercritical regime . . . 45
3.6.4 Connected regime . . . 45
3.7 Mathematical description of the giant component . . . 46
3.7.1 Interpretation of the self-consistency equation . . . 46
3.7.2 Near-threshold behavior . . . 46
3.7.3 Structure in the supercritical regime . . . 47
3.7.4 Connectedness threshold . . . 47
3.8 Numerical exploration . . . 48
3.8.1 Largest component fraction . . . 48
3.8.2 Number of connected components . . . 49
3.8.3 Fraction of isolated vertices . . . 50
3.8.4 Probability of connectedness . . . 51
3.9 Summary of regimes . . . 52
3.10 Core formulas to remember . . . 53
3.11 In-class questions . . . 53
3.12 End-of-chapter exercises . . . 53
3.12.1 Part I. Mathematical exercises . . . 53
3.12.2 Part II. Conceptual and analytical exercises . . . 55

3.12.3 Part III. Simulation and coding exercises . . . 56
3.13 Suggested project-style extension . . . 58
3.14 Concluding remarks . . . 59

4 Small-World Networks **60**
4.1 Introduction . . . 60
4.2 Learning goals . . . 60
4.3 Distance, diameter, and average path length . . . 61
4.3.1 Distance . . . 61
4.3.2 Diameter . . . 61
4.3.3 Average path length . . . 61
4.3.4 Why averaging suppresses fluctuations . . . 61
4.3.5 Clustering coefficient . . . 62
4.4 The small-world property in random networks . . . 62
4.4.1 A branching approximation . . . 62
4.4.2 Number of vertices within distance d . . . 62
4.4.3 Logarithmic estimate for distances . . . 63
4.4.4 Average path length in random networks . . . 63
4.4.5 A social-network illustration . . . 64
4.5 Why small worlds are surprising . . . 64
4.5.1 Distance scaling in regular lattices . . . 64
4.5.2 Why the contrast matters . . . 64
4.5.3 Simulated scaling of average path length . . . 65
4.6 The Watts–Strogatz model . . . 69
4.6.1 Motivation . . . 69
4.6.2 Parameters of the model . . . 69
4.6.3 Regular ring lattice . . . 69
4.6.4 Definition of the Watts–Strogatz model . . . 70
4.6.5 Why only the clockwise edges are considered . . . 70
4.6.6 Interpretation of the parameter β . . . 70
4.6.7 Ring-lattice benchmarks . . . 71
4.6.8 Visual interpolation from order to randomness . . . 72
4.6.9 Why shortcuts are so effective . . . 74
4.6.10 Quantitative transition . . . 75
4.6.11 Numerical experiment . . . 75
4.7 Small-world regime . . . 76
4.8 What the Watts–Strogatz model explains and what it does not . . . 77
4.9 Applications of small-world networks . . . 77
4.9.1 Social systems . . . 78
4.9.2 Neural systems . . . 78
4.9.3 Transportation and infrastructure . . . 78
4.9.4 Communication and information systems . . . 78
4.9.5 Why applications matter . . . 78
4.10 Core formulas to remember . . . 78
4.11 In-class questions . . . 79
4.12 End-of-chapter exercises . . . 79
4.12.1 Part I. Mathematical exercises . . . 79
4.12.2 Part II. Conceptual and analytical exercises . . . 81
4.12.3 Part III. Simulation and coding exercises . . . 82
4.12.4 Suggested mini-project . . . 83
4.13 Concluding remarks . . . 83

5 Scale-Free Networks **84**
5.1 Introduction . . . 84

5.2 Learning goals . . . 84
5.3 Broad degree distributions and hubs . . . 85
5.3.1 Degree heterogeneity . . . 85
5.3.2 Hubs . . . 85
5.3.3 Visual contrast: homogeneous and heterogeneous degree structure . . . 86
5.4 Power-law degree distributions . . . 86
5.4.1 Definition of a power law . . . 86
5.4.2 Why the term “scale-free” is used . . . 87
5.4.3 Cumulative distribution . . . 87
5.5 Preferential attachment . . . 87
5.5.1 Interpretation . . . 88
5.5.2 Why preferential attachment matters . . . 88
5.6 The Barabási–Albert model . . . 88
5.6.1 Average degree in the BA model . . . 88
5.7 Growth of a preferential-attachment network . . . 89
5.8 Degree distribution in the BA model . . . 89
5.8.1 Consequences of a heavy tail . . . 90
5.9 Numerical exploration . . . 90
5.9.1 Degree distribution of a BA network . . . 90
5.9.2 Cumulative degree distribution . . . 92
5.10 Comparison with Erdős–Rényi graphs . . . 92
5.11 Why scale-free structure is important . . . 93
5.11.1 Paths and centralization . . . 93
5.11.2 Robustness and vulnerability . . . 93
5.11.3 Spreading processes . . . 93
5.12 Core formulas to remember . . . 94
5.13 In-class questions . . . 94
5.14 End-of-chapter exercises . . . 94
5.14.1 Part I. Mathematical exercises . . . 95
5.14.2 Part II. Conceptual and analytical exercises . . . 96
5.14.3 Part III. Simulation and coding exercises . . . 97
5.14.4 Suggested mini-project . . . 98
5.15 Concluding remarks . . . 98

Chapter 1

Course Overview

1.1 Welcome

This Quarto book is designed as a single, well-structured version of a Network Science course.

Each chapter follows the same pattern:

1. **Theory and intuition**
2. **Worked examples in R**
3. **Exercises and coding tasks**

The same source files produce both:

- an HTML course website
- a PDF course handout

1.2 Required packages

Install the packages below once:

```
install.packages(c(
  "igraph", "dplyr", "ggplot2", "tidyr", "purrr", "plotly", "tinytex", "quarto", "ggraph"
))
tinytex::install_tinytex()
```

1.3 Current chapters

The current book contains these chapters:

- Random Graphs
- Random Graph Evolution
- Small-World Graphs
- Scale-free Graphs

Chapter 2

Random Graphs

2.1 Introduction

Random graphs are one of the most important starting points in network science. In ordinary graph theory, the usual starting point is a fixed graph: the nodes and links are already given, and the goal is to analyse the structure that is already there. In random graph theory, the viewpoint changes. Instead of starting from one fixed graph, the starting point is a rule for generating graphs.

This change is conceptually important. It means that the object of interest is not one network, but a whole collection of possible networks that could be produced by the same random mechanism. Some of these graphs are more likely than others, and many questions are no longer about one exact object. Instead, the natural questions become:

- What is the expected number of links?
- What is the expected average degree?
- What does the degree distribution typically look like?
- How likely is it that triangles or large connected pieces appear?

This is why random graphs are such a useful first model. They provide a clean probabilistic baseline and a simple reference point for later network models.

2.2 Learning goals

By the end of this chapter, it should be possible to:

- explain what a graph ensemble is
- distinguish between the models $G(N, L)$ and $G(N, p)$
- compute expected average degree
- derive the degree distribution in $G(N, p)$
- explain why a Poisson approximation appears in sparse large graphs
- define clustering and interpret what it means
- connect the formulas to simulations in `R`

2.3 A beginner-friendly motivation

A simple way to build intuition is to think about a classroom during the first week of a semester.

Suppose there are N students in a class. At the beginning, not everyone knows everyone else. Over the first few days, pairs of students may talk, exchange contact information, or work together in small activities. If a graph is drawn in which:

- each student is a node
- a link appears when two students become acquainted

then the resulting network depends on chance. Some pairs talk, some do not. Some classes become highly connected, while others remain sparse.

Now imagine that this process is repeated many times with different classes of the same size and under roughly similar conditions. The exact same acquaintance network should not be expected every time. But certain patterns may still be stable:

- the average number of acquaintances per student may be similar
- the probability of observing a student with degree k may follow a predictable distribution
- the network may be denser when the chance of interaction is higher

This is the core idea of a random graph model. The claim is not that real social interaction is random in every detail. Rather, randomness is used as a baseline model. If a simple random rule already explains some feature, that is informative. If a real network differs strongly from the random baseline, that is informative too.

> **Main idea**
>
> A random graph is not a single graph. It is a probability model that generates many possible graphs.

2.4 Why the binomial distribution appears

Before defining the Erdős–Rényi models, it helps to recall the binomial distribution, because it appears naturally in this chapter.

If a random variable X counts the number of successes in n independent trials, where each trial succeeds with probability p, then

$$X \sim B(n, p)$$

and

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Its mean is

$$\mathbb{E}[X] = np$$

and its variance is

$$\mathrm{Var}(X) = np(1-p).$$

This matters because in the model $G(N, p)$, every possible edge is treated as an independent trial:

- success = the edge is present
- failure = the edge is absent

As a result, both the number of edges and the degree of a node can be described using binomial ideas.

2.5 The Erdős–Rényi models

There are two classical random graph models that are closely related but conceptually different.

2.5.1 The model $G(N, L)$

In the model $G(N, L)$:

- the number of nodes is fixed and equal to N
- the number of links is fixed and equal to L
- all simple graphs with exactly those values are equally likely

The word *simple* means that self-loops and multiple edges are excluded.

If there are N nodes, then the total number of possible undirected edges is

$$K = \binom{N}{2} = \frac{N(N-1)}{2}.$$

To construct a graph with exactly L links, L edges must be chosen out of these K possible ones. Therefore, the number of distinct graphs in this ensemble is

$$\Omega(N, L) = \binom{K}{L}.$$

Since each such graph is equally likely, the probability of any one graph G is

$$P(G) = \frac{1}{\binom{K}{L}}.$$

2.5.1.1 Intuition for $G(N, L)$

This model keeps the density tightly controlled because the final graph always has exactly L links. The randomness lies only in **where** those links are placed.

So $G(N, L)$ is useful when all graphs with the same size and the same number of links are being compared.

2.5.1.2 Example: the ensemble $G(3, 2)$

Take three labeled nodes: 1, 2, and 3.

For $N = 3$, the total number of possible undirected edges is

$$K = \binom{3}{2} = 3.$$

These three possible edges are

$$(1, 2), \qquad (1, 3), \qquad (2, 3).$$

Now consider the model $G(3, 2)$. Since exactly two edges must be present, the number of possible labeled graphs is

$$\Omega(3, 2) = \binom{3}{2} = 3.$$

So there are exactly three distinct labeled graphs in this ensemble, and each one has probability

$$P(G) = \frac{1}{3}.$$

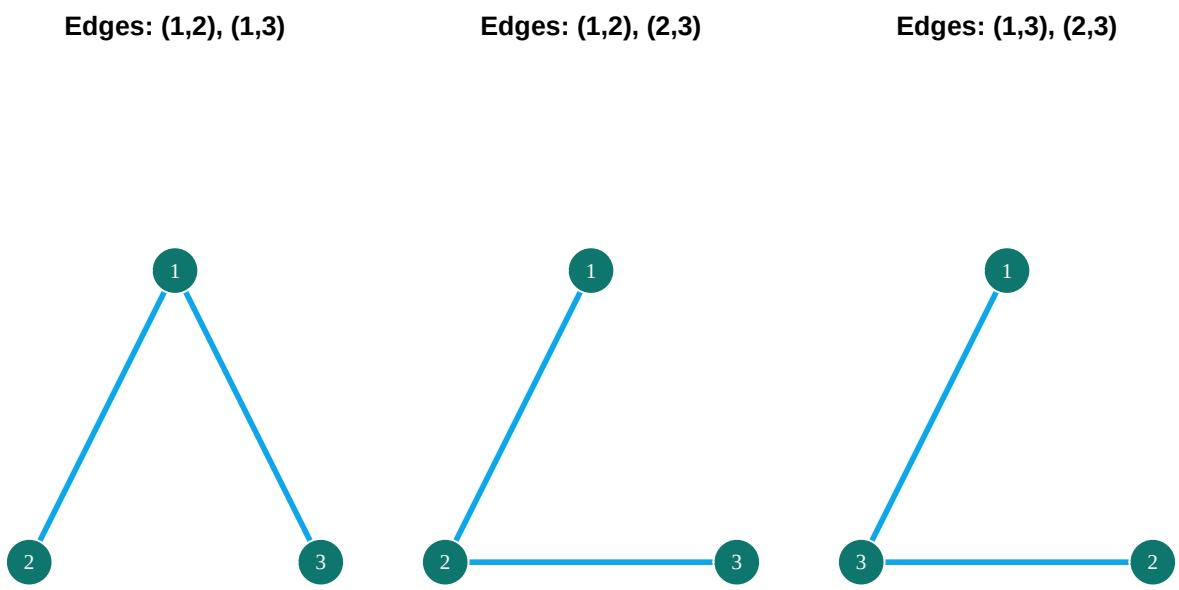


Figure 2.1: The three labeled graphs in the ensemble G(3,2).

This example makes the structure of $G(N, L)$ very concrete:

- the node set is fixed
- the number of edges is fixed
- the only randomness is which L edges are chosen

2.5.2 The model $G(N, p)$

In the model $G(N, p)$:

- the number of nodes is fixed and equal to N
- each possible edge is included independently with probability p

Again, the number of possible undirected edges is

$$K = \binom{N}{2}.$$

Now suppose a generated graph ends up with exactly L links. Then:

- each present edge contributes a factor of p
- each absent edge contributes a factor of $(1 - p)$

Since there are L present edges and $K - L$ absent edges, the probability of that graph is

$$P(G) = p^L (1-p)^{K-L}.$$

2.5.2.1 Intuition for $G(N, p)$

This model does **not** fix the total number of links in advance. Instead, every possible pair of nodes gets its own independent chance to be connected.

That makes $G(N, p)$ especially natural for situations like:

- random failure of communication links
- random activation of pathways
- random acquaintance formation in a large group
- a baseline model where every pair has the same connection probability

2.5.2.2 Example: the ensemble $G(3, p)$

Again take three labeled nodes: 1, 2, and 3.

There are still exactly three possible edges:

$$(1, 2), \qquad (1, 3), \qquad (2, 3).$$

In the model $G(3, p)$, each of these three edges may be either:

- present, with probability p
- absent, with probability $1 - p$

Since each edge has two possibilities and the three edge decisions are independent, the total number of distinct labeled graphs is

$$2^3 = 8.$$

So the ensemble $G(3, p)$ contains eight labeled graphs.

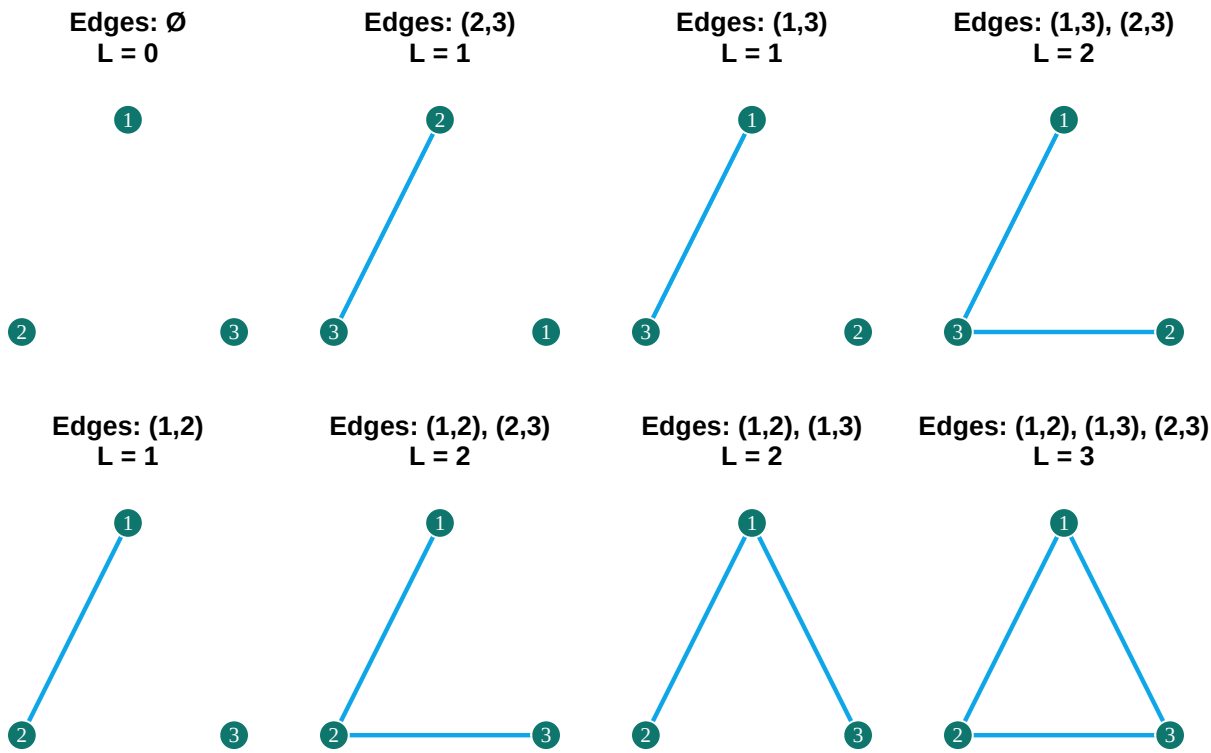


Figure 2.2: The eight labeled graphs in the ensemble G(3,p).

In this case, the number of edges is not fixed. Some of the eight graphs have:

- $L = 0$ edges
- $L = 1$ edge
- $L = 2$ edges
- $L = 3$ edges

What matters is that each possible edge is decided independently.

2.5.2.3 Probability table for $G(3, p)$

Since each graph is determined by the presence or absence of the three possible edges, its probability depends only on how many edges it contains.

Graph	Present edges	Number of edges L	Probability
1	$\emptyset$	0	$(1-p)^3$
2	$(1,2)$	1	$p(1-p)^2$
3	$(1,3)$	1	$p(1-p)^2$
4	$(2,3)$	1	$p(1-p)^2$
5	$(1,2),(1,3)$	2	$p^2(1-p)$
6	$(1,2),(2,3)$	2	$p^2(1-p)$
7	$(1,3),(2,3)$	2	$p^2(1-p)$
8	$(1,2),(1,3),(2,3)$	3	p^3

This shows that in $G(3, p)$ all eight labeled graphs are allowed, but they are not equally likely. Their probabilities depend on how many edges are present.

2.5.3 Comparing $G(N, L)$ and $G(N, p)$

The two models are often confused at first, so it is worth separating them clearly.

- In $G(N, L)$, the number of edges is fixed.
- In $G(N, p)$, the number of edges is random.
- In $G(N, L)$, all allowed graphs are equally likely.
- In $G(N, p)$, graphs with the same number of links have the same probability, but graphs with different numbers of links generally do not.

> **!** Conceptual difference
>
> The model $G(N, L)$ is a uniform choice among graphs with exactly L edges.
> The model $G(N, p)$ is an independent decision process for each possible edge.

2.6 Average degree

Average degree is one of the first global quantities used to summarize a network. It gives a single number that describes, on average, how many neighbors each node has.

If a graph has N nodes and L links, then the average degree is

$$\bar{k} = \frac{1}{N} \sum_{i=1}^{N} k_i,$$

where k_i is the degree of node i.

Using the handshake identity,

$$\sum_{i=1}^{N} k_i = 2L,$$

it follows that

$$\bar{k} = \frac{2L}{N}.$$

This formula is fundamental. It says that the average degree is determined entirely by the number of nodes and the number of links.

A useful interpretation is the following:

- each link contributes 1 to the degree of one endpoint
- and also contributes 1 to the degree of the other endpoint
- so each link contributes a total of 2 to the sum of all node degrees

That is why the numerator is $2L$ rather than L.

2.6.1 A quick numerical example

Suppose a graph has

$$N = 10 \qquad \text{and} \qquad L = 15.$$

Then the average degree is

$$\bar{k} = \frac{2L}{N} = \frac{2 \times 15}{10} = 3.$$

This does **not** mean that every node has degree 3. It only means that the average over all node degrees is 3.

For example, one graph could have degrees

$$(1, 2, 2, 3, 3, 3, 4, 4, 4, 4),$$

whose average is 3. Another graph with a very different structure might still have the same average degree. So average degree is useful, but it does not capture the full shape of the degree distribution.

2.6.2 Why average degree matters in random graphs

In a fixed graph, $\bar{k}$ is a property of that one graph. In a random graph model, the situation is different because the graph itself is random. That means the number of links may vary from one realization to another, and therefore the average degree may also vary.

So in a random graph ensemble, the natural quantity to study is the **expected average degree**,

$$\mathbb{E}[\bar{k}].$$

This is the average value of $\bar{k}$ over all graphs generated by the model.

Starting from

$$\bar{k} = \frac{2L}{N},$$

and taking expectation on both sides gives

$$\mathbb{E}[\bar{k}] = \mathbb{E}\left[\frac{2L}{N}\right].$$

Since N is fixed in both Erdős–Rényi models, the constant $\frac{2}{N}$ can be taken outside the expectation:

$$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N}.$$

This relation is extremely useful. It says that once the expected number of links is known, the expected average degree follows immediately.

> **Key relation**
>
> For any random graph ensemble with fixed number of nodes N,
>
> $$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N}.$$
>
> So the problem of expected average degree reduces to the problem of expected number of links.

2.6.3 Average degree in $G(N, L)$

In the model $G(N, L)$, the number of links is fixed from the beginning. Every graph in the ensemble has exactly L links.

That means there is no randomness in the value of L itself:

$$\mathbb{E}[L] = L.$$

Substituting this into the general relation gives

$$\mathbb{E}[\bar{k}] = \frac{2L}{N}.$$

So in $G(N, L)$, the average degree is not only expected to equal $\frac{2L}{N}$; in fact every graph in the ensemble has that same average degree.

This is an important distinction. The detailed arrangement of edges changes from one graph to another, but the total number of edges does not. Therefore, the average degree is fixed across the whole ensemble.

2.6.3.1 Interpretation in $G(N, L)$

The model $G(N, L)$ allows randomness in **structure**, but not in **density**.

- which nodes are connected is random
- how many total links exist is fixed
- therefore the average degree is fixed

For example, in $G(6, 4)$ the average degree is always

$$\bar{k} = \frac{2 \times 4}{6} = \frac{8}{6} = \frac{4}{3}.$$

Different realizations may have very different shapes, but all of them must share this same average degree.

2.6.4 Average degree in $G(N, p)$

The model $G(N, p)$ is different because the total number of links is no longer fixed in advance.

There are

$$K = \binom{N}{2}$$

possible undirected edges, and each one is included independently with probability p.

So the total number of links L is a binomial random variable:

$$L \sim B(K, p).$$

This immediately gives

$$\mathbb{E}[L] = pK.$$

Substituting into the general relation,

$$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N} = \frac{2pK}{N}.$$

Now replace K by $\binom{N}{2}$:

$$\mathbb{E}[\bar{k}] = \frac{2p}{N}\binom{N}{2}.$$

Since

$$\binom{N}{2} = \frac{N(N-1)}{2},$$

this simplifies to

$$\mathbb{E}[\bar{k}] = \frac{2p}{N} \cdot \frac{N(N-1)}{2} = p(N-1).$$

So the expected average degree in $G(N, p)$ is

$$\mathbb{E}[\bar{k}] = p(N-1).$$

2.6.5 A second way to understand the formula

The formula

$$\mathbb{E}[\bar{k}] = p(N-1)$$

can also be understood directly, without first going through $\mathbb{E}[L]$.

Take one node. It has $N-1$ possible neighbors. For each of those possible neighbors, an edge is present with probability p.

So the expected number of neighbors of that node is simply

$$(N-1) \times p = p(N-1).$$

Because all nodes are statistically equivalent in the Erdős–Rényi model, this is also the expected average degree of the whole graph.

This second interpretation is very useful because it makes the formula intuitive:

- $N-1$ = number of possible neighbors
- p = chance that each possible link exists
- expected degree = "number of opportunities" times "probability of success"

2.6.6 A quick numerical example in $G(N, p)$

Suppose

$$N = 50 \qquad \text{and} \qquad p = 0.10.$$

Then

$$\mathbb{E}[\bar{k}] = p(N-1) = 0.10 \times 49 = 4.9.$$

So a typical graph in this ensemble should have an average degree close to 4.9, although the exact value will vary from one realization to another.

This illustrates an important contrast with $G(N, L)$:

- in $G(N, L)$ the average degree is exactly fixed
- in $G(N, p)$ the average degree fluctuates, but its expected value is fixed

2.6.7 Dense and sparse regimes

The formula

$$\mathbb{E}[\bar{k}] = p(N-1)$$

also helps explain how the graph changes when N or p changes.

If p is fixed and N increases, then the expected average degree grows roughly linearly with N. In that case the graph becomes denser as the system grows.

On the other hand, many real network settings are better modeled by keeping the expected degree moderate even when N becomes large. To do that, p must decrease as N grows. This is exactly the sparse regime that later leads to the Poisson approximation for the degree distribution.

So the average degree is not just a descriptive quantity. It also helps identify whether the graph is sparse or dense.

2.6.8 Comparing the two models

The formulas for the two Erdős–Rényi models are:

$$\mathbb{E}[\bar{k}] = \frac{2L}{N} \qquad \text{for } G(N, L),$$

and

$$\mathbb{E}[\bar{k}] = p(N-1) \qquad \text{for } G(N, p).$$

These two expressions describe the same idea in two different parameterizations:

- in $G(N, L)$ the model is controlled by the number of links L
- in $G(N, p)$ the model is controlled by the edge probability p

They are closely related. In fact, if a graph from $G(N, p)$ has expected number of links

$$\mathbb{E}[L] = p\binom{N}{2},$$

then it is natural to compare it with a graph from $G(N, L)$ using approximately that same value of L.

2.6.9 Summary

The average degree is one of the simplest and most useful global quantities in random graph theory.

The main points are:

- in any graph with N nodes and L links,
$$\bar{k} = \frac{2L}{N}$$
- in a random graph ensemble,
$$\mathbb{E}[\bar{k}] = \frac{2\,\mathbb{E}[L]}{N}$$
- in $G(N, L)$,
$$\mathbb{E}[\bar{k}] = \frac{2L}{N}$$
- in $G(N, p)$,
$$\mathbb{E}[\bar{k}] = p(N-1)$$

So the expected average degree gives a first measure of how densely connected a random graph is, and it provides a natural bridge between the combinatorial description of the model and its probabilistic behavior.

2.7 Degree distribution

The degree distribution describes how node degrees are distributed across a graph. In this chapter, the important point is not the definition of degree itself, but how a random graph model gives rise to a distribution of degrees across nodes.

A useful way to introduce this idea is to move in three steps:

1. start from one concrete graph
2. record the degree of each node
3. transform those degree values into a distribution

2.7.1 A small example

Consider the following graph.

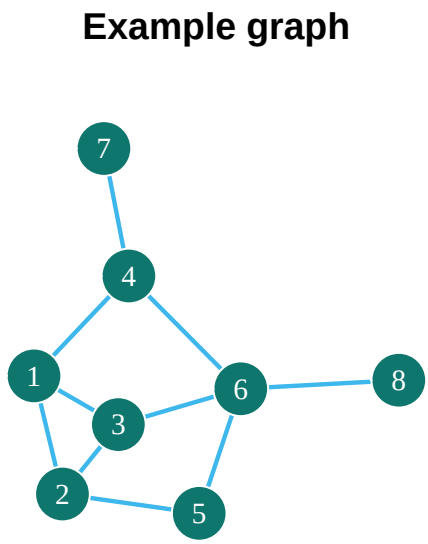


Figure 2.3: A small graph used to build the degree distribution step by step.

For this graph, the degree of each node is:

Node	Degree
1	3
2	3
3	3
4	3
5	2
6	4
7	1
8	1

This table contains node-level information. It tells the degree of each specific node, but it does not yet summarize how the degree values are distributed across the graph.

2.7.2 From node degrees to degree counts

The next step is to count how many nodes share each degree value.

Degree k	Number of nodes N_k	Proportion $p_k = \frac{N_k}{N}$
0	0	0
1	2	0.25
2	1	0.125

Degree k	Number of nodes N_k	Proportion $p_k = \frac{N_k}{N}$
3	4	0.50
4	1	0.125

This table is the key transition from node-level data to a graph-level summary.

- the first column gives the degree value k
- the second column gives the number of nodes having that degree
- the third column gives the fraction of nodes having that degree

If the graph has N nodes in total and N_k nodes of degree k, then the empirical degree distribution is

$$p_k = \frac{N_k}{N}.$$

So p_k is the proportion of nodes whose degree is exactly k.

2.7.3 Plotting the degree distribution

The same information can be shown visually.

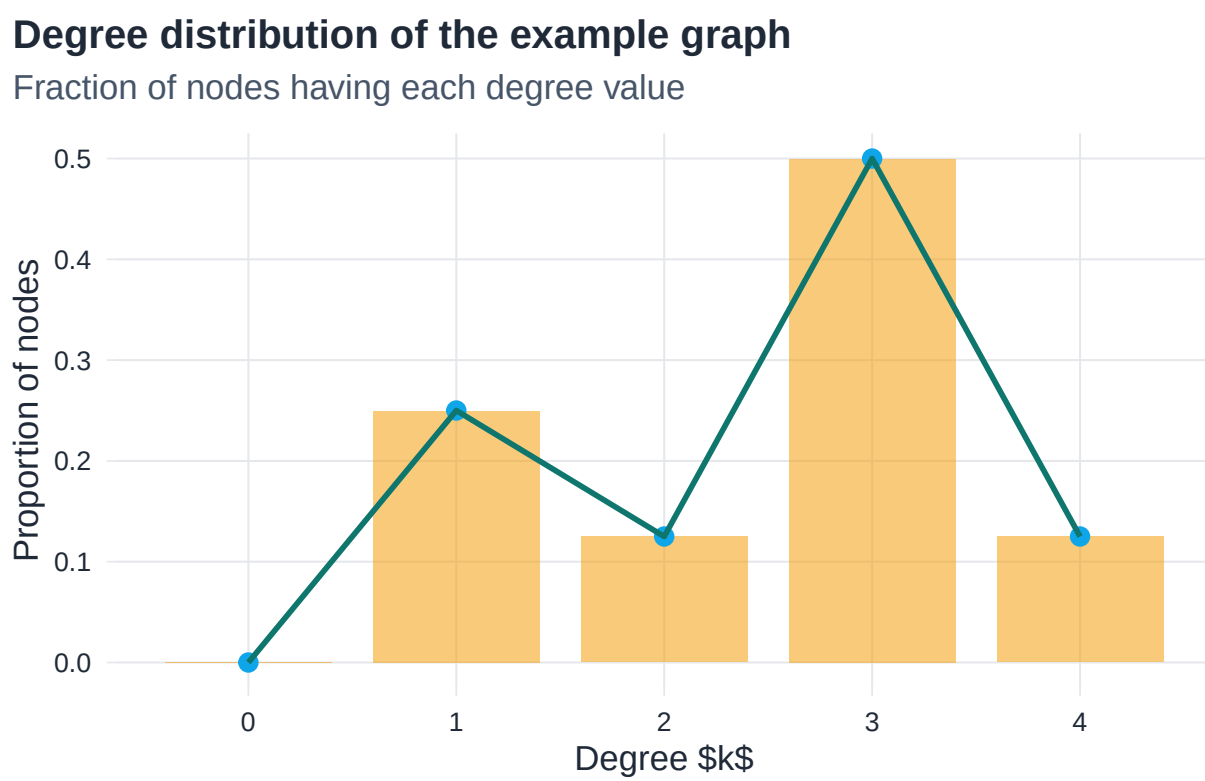


Figure 2.4: Degree distribution of the example graph.

This plot shows the degree distribution in its most direct form:

- the horizontal axis gives the degree value k
- the vertical axis gives the proportion of nodes with that degree

The distinction between two ideas is now clear:

- the **degree table** records the degree of each node
- the **degree distribution** records how common each degree value is in the graph

2.7.4 Degree distribution in the random graph model $G(N, p)$

Now move from one fixed graph to the random graph model $G(N, p)$.

Take one node. It has $N-1$ possible neighbors. For each possible neighbor, an edge is present with probability p and absent with probability $1-p$. Because these edge decisions are independent, the degree of that node is the number of successes in $N-1$ Bernoulli trials. Therefore,

$$k \sim B(N-1, p).$$

Hence the probability that a node has degree exactly k is

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$

This is the degree distribution of the Erdős–Rényi model $G(N, p)$.

2.7.5 Interpreting the formula

Each factor in the formula has a clear meaning.

To obtain degree k for a chosen node:

1. exactly k of the $N-1$ possible edges must be present
2. the remaining $N-1-k$ edges must be absent
3. there are $\binom{N-1}{k}$ different ways to choose which k neighbors are connected

Therefore,

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$

The three parts of this expression play different roles:

- $\binom{N-1}{k}$ counts the possible choices of the k neighbors
- p^k gives the probability that those k edges are present
- $(1-p)^{N-1-k}$ gives the probability that the remaining edges are absent

The expected value of this distribution is

$$\mathbb{E}[k] = p(N-1),$$

which agrees with the expected average degree derived earlier.

2.7.6 Comparing one realization with the theoretical distribution

The formula above describes the degree distribution predicted by the model. A single simulated graph from $G(N, p)$ will fluctuate around that prediction.

```r
n <- 80
p <- 0.08

g <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)
obs_deg <- degree(g)

obs_tbl <- tibble(k = obs_deg) |>
  count(k, name = "observed") |>
  complete(k = 0:(n - 1), fill = list(observed = 0)) |>
  mutate(observed_prob = observed / sum(observed))

theory_tbl <- tibble(k = 0:(n - 1)) |>
  mutate(theory_prob = dbinom(k, size = n - 1, prob = p))

plot_tbl <- left_join(obs_tbl, theory_tbl, by = "k")

ggplot(plot_tbl, aes(x = k)) +
  geom_col(aes(y = observed_prob), fill = course_cols$blue, alpha = 0.5, width = 0.82) +
  geom_line(aes(y = theory_prob), color = course_cols$gold, linewidth = 1.1) +
  geom_point(aes(y = theory_prob), color = course_cols$teal, size = 2.2) +
  labs(
    title = "Observed and theoretical degree distribution",
    subtitle = sprintf("Single realization of G(%d, %.2f)", n, p),
    x = "Degree $k$",
    y = "Probability"
  )
```

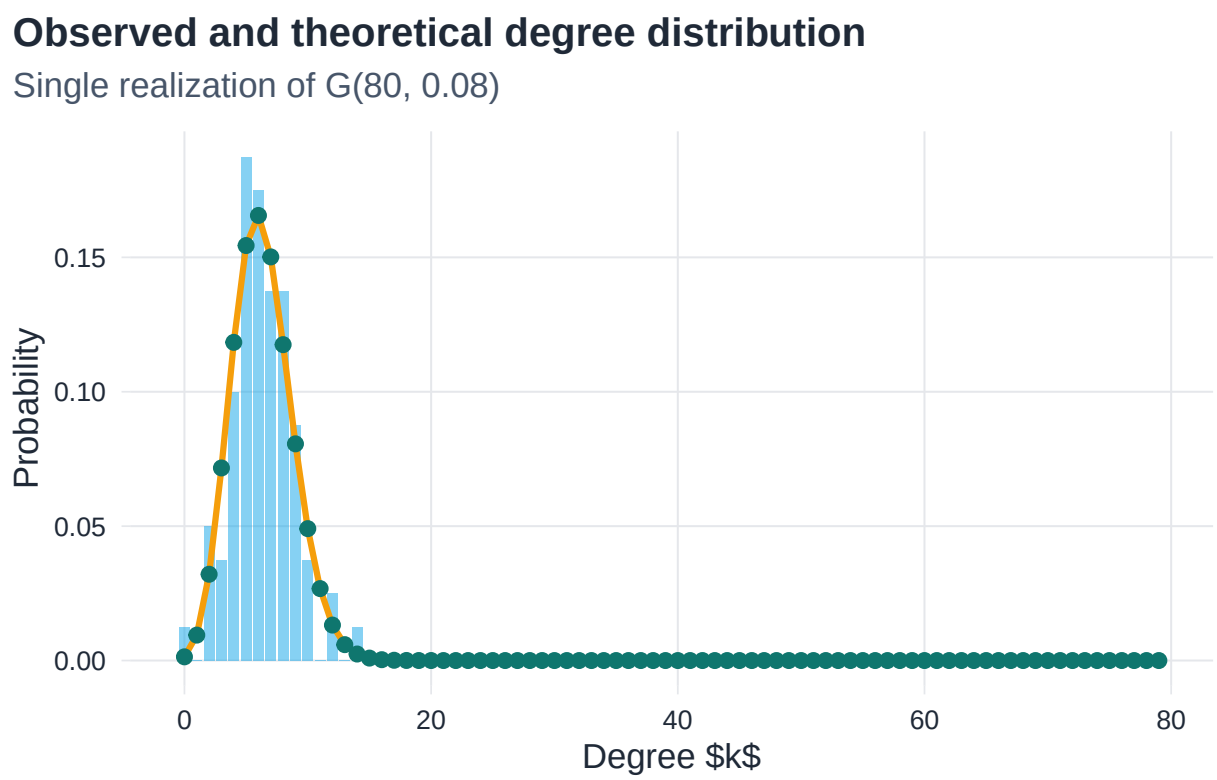


Figure 2.5: Observed degree frequencies in one simulated graph compared with the theoretical degree distribution in G(N,p).

The bars show the empirical degree distribution of one generated graph. The line and points show the theoretical distribution

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$

The two are not expected to coincide perfectly in one realization. The figure shows how the observed degree frequencies of a random graph relate to the probability law defined by the model.

! Summary

In this chapter, the degree distribution can be read at three connected levels:

1. a graph has nodes with different degree values
2. those degree values can be summarized by the empirical distribution

$$p_k = \frac{N_k}{N}$$

3. in the model $G(N, p)$, the degree of a node follows the binomial law

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

2.8 Poisson approximation in sparse large graphs

The exact binomial formula is correct, but in many network settings the focus is on large graphs that are relatively sparse.

A graph is called **sparse** when the number of links grows much more slowly than the maximum possible number of links. In the Erdős–Rényi setting, a common sparse regime is:

- N is large
- p is small
- the mean degree remains finite

Let

$$\lambda = p(N-1).$$

If N is large and p is small in such a way that λ stays finite, then the binomial degree distribution is well approximated by a Poisson distribution:

$$P(k) \approx e^{-\lambda} \frac{\lambda^k}{k!}.$$

2.8.1 Why this approximation appears

This approximation reflects a common probabilistic pattern:

- each individual edge is unlikely
- there are many possible edges
- the total number of successes remains moderate

That is exactly the setting in which Poisson limits naturally appear.

This is an important modeling lesson. The Poisson distribution is not introduced here as a completely different idea. It appears as a limit of the binomial degree distribution in the sparse large-N regime.

> Interpretation
>
> In a large sparse Erdős–Rényi graph, most nodes have relatively small degree, and the degree distribution becomes sharply concentrated around a finite mean.

2.8.2 Class example: binomial distribution and Poisson approximation

```
n <- 500
p <- 0.02
lambda <- p * (n - 1)
k_vals <- 0:24

compare_tbl <- tibble(
  k = k_vals,
  binomial = dbinom(k_vals, size = n - 1, prob = p),
  poisson = dpois(k_vals, lambda = lambda)
) |>
  pivot_longer(cols = c(binomial, poisson), names_to = "distribution", values_to = "probability") |>
  mutate(distribution = recode(distribution, binomial = "Binomial", poisson = "Poisson"))

ggplot(compare_tbl, aes(x = k, y = probability, color = distribution)) +
```

```
  geom_line(linewidth = 1.1) +
  geom_point(size = 2.2) +
  scale_color_manual(values = c("Binomial" = course_cols$gold, "Poisson" = course_cols$teal)) +
  labs(
    title = "Binomial and Poisson degree distributions",
    subtitle = sprintf("n = %d, p = %.3f, mean degree = %.2f", n, p, lambda),
    x = "Degree $k$",
    y = "Probability",
    color = NULL
  )
```

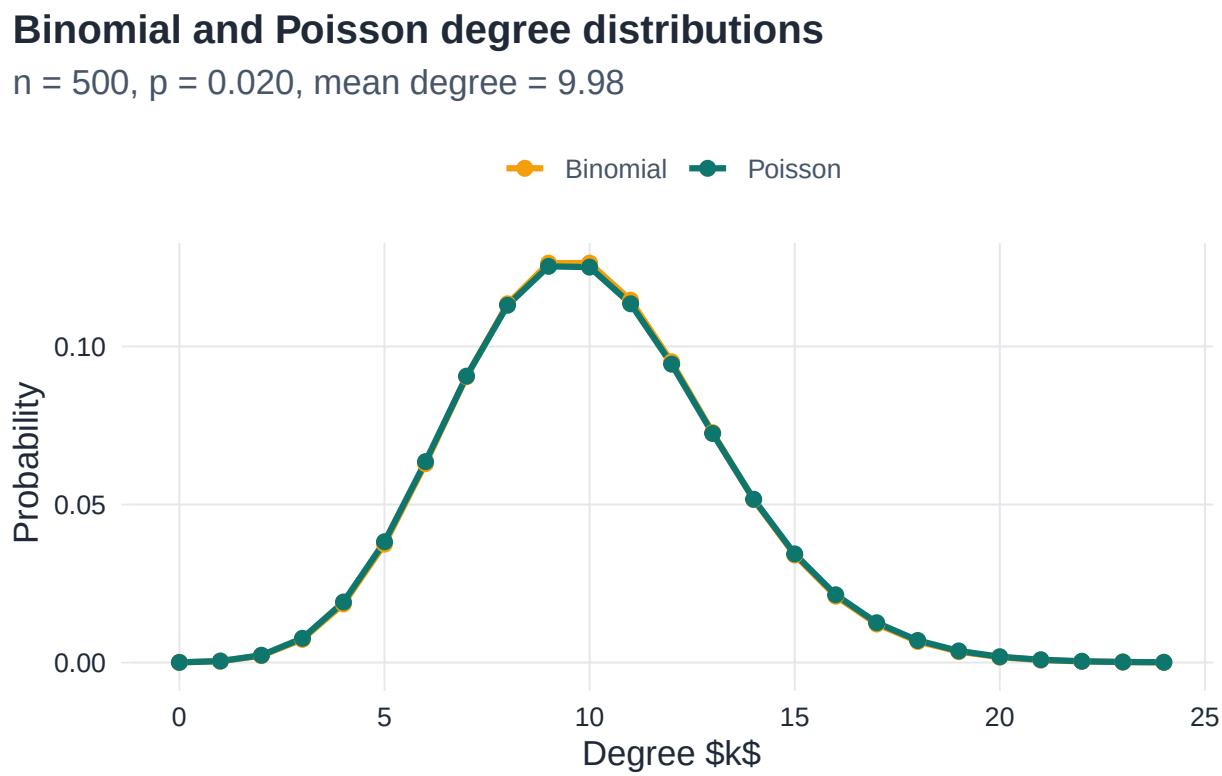


Figure 2.6: Comparison between the exact binomial degree distribution and its Poisson approximation.

2.9 Clustering coefficient

The clustering coefficient measures how strongly the neighbors of a node are connected to one another. In other words, once a node is chosen, clustering asks whether its neighbors also tend to be neighbors of each other.

This idea is important because two graphs can have similar size, similar average degree, and even similar degree distributions, while still differing strongly in how much local triangle structure they contain. The clustering coefficient is one of the main quantities used to capture that local structure.

2.9.1 Local clustering coefficient

Suppose node i has degree k_i. Then node i has exactly k_i neighbors.

Among those k_i neighbors, the maximum possible number of edges is

$$L_i^{\max} = \binom{k_i}{2} = \frac{k_i(k_i-1)}{2}.$$

If the actual number of edges among those neighbors is L_i, then the local clustering coefficient of node i is

$$C_i = \frac{L_i}{L_i^{\max}} = \frac{2L_i}{k_i(k_i-1)}.$$

This formula compares:

- the number of neighbor-neighbor edges that actually exist
- the maximum number of neighbor-neighbor edges that could exist

So C_i is always between 0 and 1:

- $C_i = 0$ means none of the neighbors of i are connected to each other
- $C_i = 1$ means all neighbors of i are mutually connected
- intermediate values indicate partial local connectivity

2.9.2 A fixed node with three different neighborhood structures

To see how this works clearly, keep the same central node A and the same three neighbors B, C, and D, and only change the edges among the neighbors.

In all three examples below:

- the focal node is A
- the neighbors of A are B, C, and D
- therefore $k_A = 3$
- the maximum possible number of edges among the neighbors is

$$L_A^{\max} = \binom{3}{2} = 3$$

So the only question is: how many of the three possible neighbor-neighbor edges actually exist?

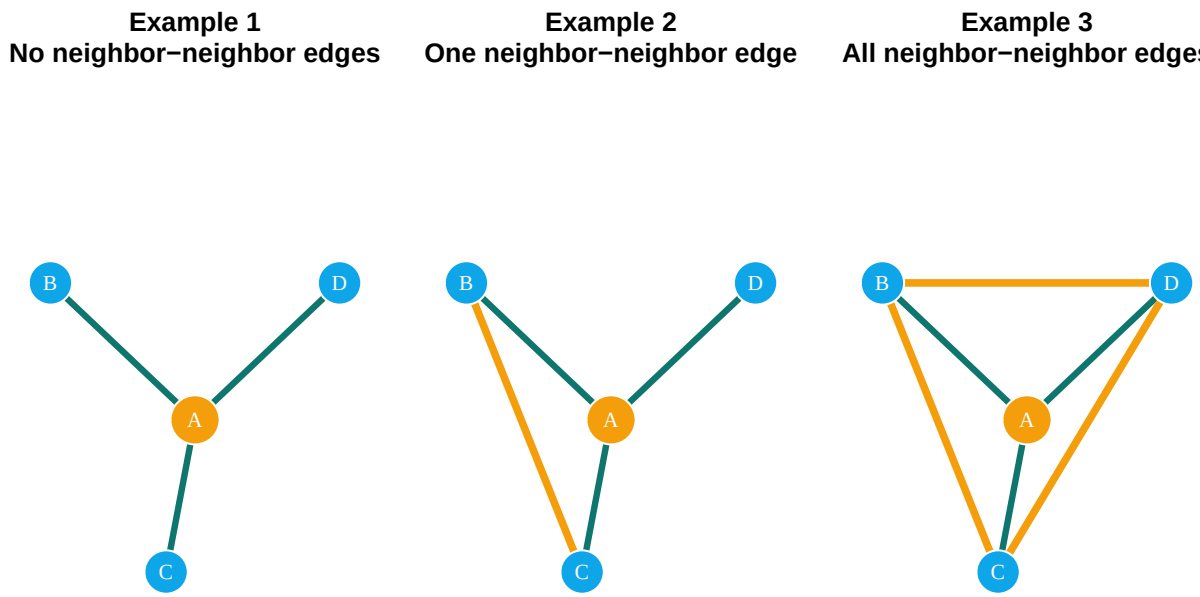


Figure 2.7: Three example neighborhoods for the same focal node A. dark green edges connect A to its neighbors; light green edges connect neighbors to each other and contribute to clustering.

In these figures:

- the **dark green edges** connect the focal node A to its neighbors
- the **light green edges** connect neighbors to each other
- only the light green edges count in the value of L_A

2.9.3 Example 1

In the first graph, node A has neighbors B, C, and D, but none of those neighbors are connected to each other.

So

$$L_A = 0 \qquad \text{and} \qquad L_A^{\max} = 3.$$

Therefore,

$$C_A = \frac{L_A}{L_A^{\max}} = \frac{0}{3} = 0.$$

This is the smallest possible clustering value. The neighbors of A form no local triangle structure at all.

2.9.4 Example 2

In the second graph, the same three neighbors are present, but now one edge exists among them, namely between B and C.

So

$$L_A = 1 \qquad \text{and} \qquad L_A^{\max} = 3.$$

Therefore,

$$C_A = \frac{L_A}{L_A^{\max}} = \frac{1}{3}.$$

This means that only one-third of the possible neighbor-neighbor edges are present.

2.9.5 Example 3

In the third graph, all three neighbor-neighbor edges are present:

$$(B, C), \qquad (B, D), \qquad (C, D).$$

So

$$L_A = 3 \qquad \text{and} \qquad L_A^{\max} = 3.$$

Therefore,

$$C_A = \frac{L_A}{L_A^{\max}} = \frac{3}{3} = 1.$$

This is the largest possible clustering value. The neighbors of A form a complete subgraph, so the local neighborhood is maximally clustered.

2.9.6 Summary table for the three examples

Example	Degree of focal node k_A	Maximum possible neighbor-neighbor edges $L_A^{\max}$	Actual neighbor-neighbor edges L_A	Clustering coefficient C_A
1	3	3	0	0
2	3	3	1	$\frac{1}{3}$
3	3	3	3	1

This table makes the logic of the formula very explicit. The degree stays the same in all three examples. What changes is only the number of links among the neighbors.

2.9.7 What clustering is really measuring

These examples show an important point:

- degree counts how many neighbors a node has
- clustering measures how tightly those neighbors are connected to each other

So two nodes can have the same degree but very different clustering coefficients.

For example, in all three graphs above, node A has degree 3. But its clustering coefficient changes from

$$0 \quad \text{to} \quad \frac{1}{3} \quad \text{to} \quad 1.$$

So clustering reveals local structural information that degree alone cannot capture.

2.9.8 Average clustering coefficient of a graph

Once the local clustering coefficient has been defined for every node, the average clustering coefficient of the graph is

$$\bar{C} = \frac{1}{N} \sum_{i=1}^{N} C_i.$$

This quantity measures the overall tendency of the graph to form locally connected neighborhoods.

It should be interpreted carefully:

- C_i is a node-level quantity
- $\bar{C}$ is a graph-level average

The local quantity tells how clustered one neighborhood is. The average quantity summarizes the graph as a whole.

2.9.9 Why clustering is important in network science

Clustering is especially important because many real networks are rich in triangle structure.

In social networks, for example, if one person knows two others, there is often an elevated chance that those two people also know each other. This kind of local closure is one of the characteristic patterns of social graphs.

Random graphs of the Erdős–Rényi type behave differently. Since edges are placed independently, there is no special mechanism that encourages triangle closure beyond the baseline edge probability itself.

That is why clustering becomes an important comparison point between random graphs and real-world networks.

2.9.10 Expected clustering in $G(N, p)$

Now consider the model $G(N, p)$.

Suppose node i has some set of neighbors. To compute its clustering coefficient, the question is whether pairs among those neighbors are connected.

In the Erdős–Rényi model, every possible edge appears independently with probability p. Therefore, once two neighbors of node i are chosen, the probability that they are connected is simply p.

So the expected fraction of connected neighbor-neighbor pairs is

$$\mathbb{E}[C_i] = p.$$

As a consequence, the expected average clustering coefficient is also

$$\mathbb{E}[\bar{C}] = p.$$

This is a very simple and very important result.

> **Key result for Erdős–Rényi graphs**
>
> In the model $G(N, p)$, the expected clustering coefficient is
>
> $$\mathbb{E}[C] = p.$$
>
> So clustering is directly controlled by the edge probability.

2.9.11 Why this result matters

This formula shows that Erdős–Rényi graphs usually have low clustering when p is small.

That is one of the main reasons why Erdős–Rényi graphs are useful as a baseline but often unrealistic as full models of social networks:

- they capture random edge formation well
- but they do not create unusually strong local triangle closure

So if a real network has clustering much larger than p, that indicates structure beyond independent random linking.

2.9.12 Numerical check by simulation

The theoretical prediction in $G(N, p)$ is

$$\mathbb{E}[\bar{C}] = p.$$

This can be checked numerically by simulating many graphs and computing the average clustering coefficient in each one.

```r
n <- 100
p <- 0.07
reps <- 250

sim_tbl <- map_dfr(seq_len(reps), function(i) {
  g <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)
  tibble(
    sim = i,
    avg_degree = mean(degree(g)),
    avg_clustering = transitivity(g, type = "average")
  )
})

summary_tbl <- tibble(
  quantity = c("Average degree", "Average clustering"),
  empirical_mean = c(mean(sim_tbl$avg_degree), mean(sim_tbl$avg_clustering)),
```

```r
  theoretical_value = c(p * (n - 1), p)
)

summary_tbl
```

```
# A tibble: 2 x 3
  quantity           empirical_mean theoretical_value
  <chr>                       <dbl>             <dbl>
1 Average degree             6.93               6.93
2 Average clustering         0.0706             0.07
```

```r
ggplot(sim_tbl, aes(x = avg_clustering)) +
  geom_histogram(bins = 24, fill = course_cols$gold, alpha = 0.65, color = "white") +
  geom_vline(xintercept = p, color = course_cols$teal, linewidth = 1.1, linetype = 2) +
  labs(
    title = "Average clustering across simulations",
    subtitle = sprintf("Dashed line = theoretical value p = %.2f", p),
    x = "Average clustering",
    y = "Count"
  )
```

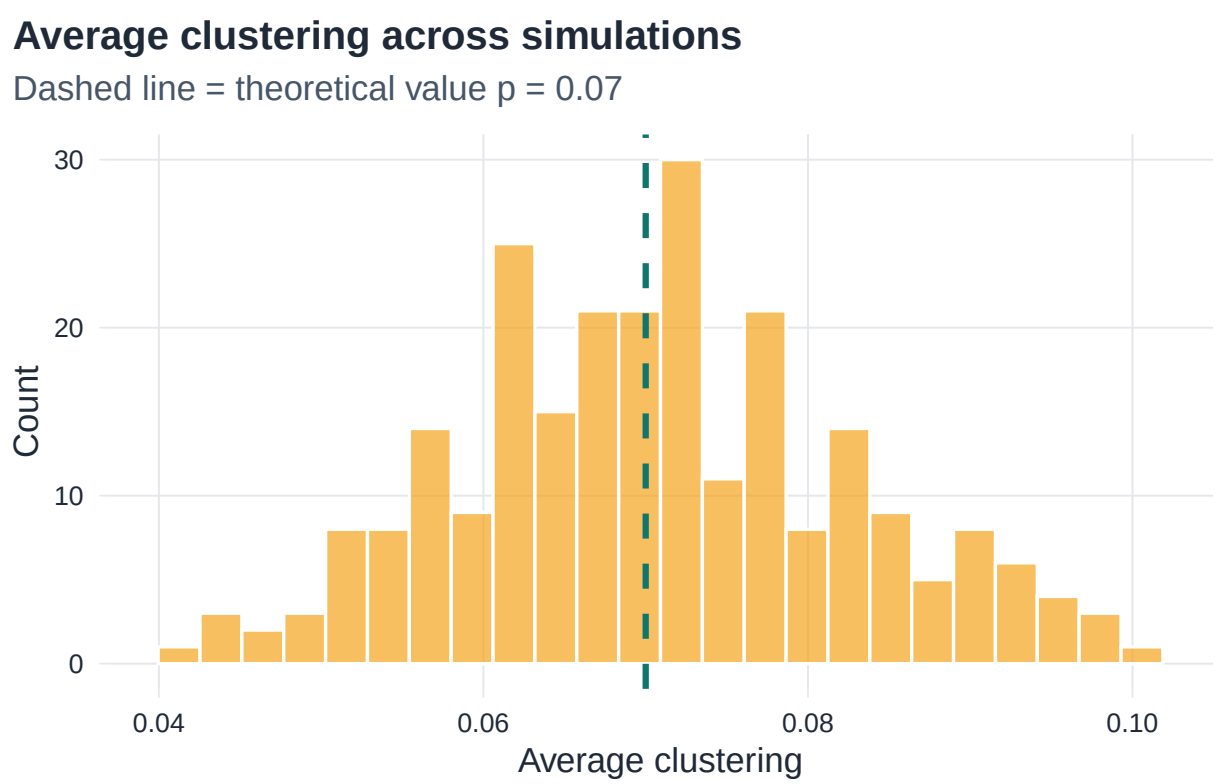


Figure 2.8: Average clustering across repeated simulations with the theoretical value shown as a dashed line.

The histogram shows how the observed average clustering varies across realizations of the model, while the dashed line marks the theoretical value p.

2.9.13 Summary

The clustering coefficient adds information that degree alone cannot provide.

The main points are:

- for a node i with degree k_i, the maximum number of possible edges among its neighbors is
$$\binom{k_i}{2}$$
- the local clustering coefficient is
$$C_i = \frac{2L_i}{k_i(k_i-1)}$$
- it measures how strongly the neighbors of a node are connected to one another
- the graph-level average is
$$\bar{C} = \frac{1}{N}\sum_{i=1}^{N} C_i$$
- in the Erdős–Rényi model $G(N,p)$,
$$\mathbb{E}[C] = p$$

So clustering is a local structural measure that becomes especially useful when comparing random graphs with more realistic network models.

2.10 Core formulas to remember

Summary

For Erdős–Rényi random graphs:

1. Number of possible undirected edges:
$$K = \binom{N}{2}$$
2. Probability of a graph in $G(N,L)$:
$$P(G) = \frac{1}{\binom{K}{L}}$$
3. Probability of a graph with L links in $G(N,p)$:
$$P(G) = p^L(1-p)^{K-L}$$
4. Expected average degree in $G(N,p)$:
$$\mathbb{E}[\bar{k}] = p(N-1)$$
5. Degree distribution in $G(N,p)$:
$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$
6. Poisson approximation in the sparse regime:
$$P(k) \approx e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \lambda = p(N-1)$$
7. Expected clustering in $G(N,p)$:
$$\mathbb{E}[C] = p$$

2.11 Class example: generate and plot a random graph

```
n <- 20
p <- 0.12

g_er <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)

plot(
  g_er,
  layout = layout_with_fr(g_er),
  vertex.size = 22,
  vertex.color = course_cols$teal,
  vertex.frame.color = "white",
  vertex.label.color = "white",
  edge.color = scales::alpha(course_cols$ink, 0.32),
  vertex.label.cex = 0.8,
  main = sprintf("A realization of G(%d, %.2f)", n, p)
)
```

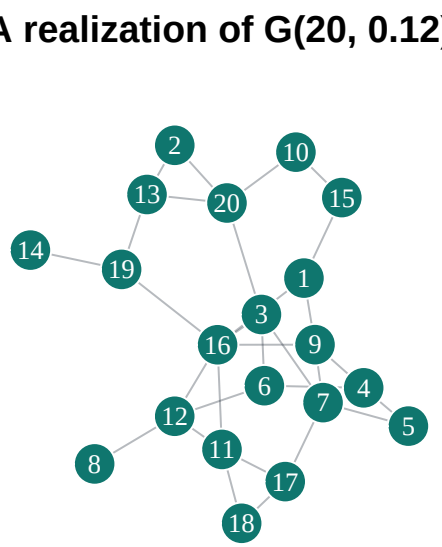
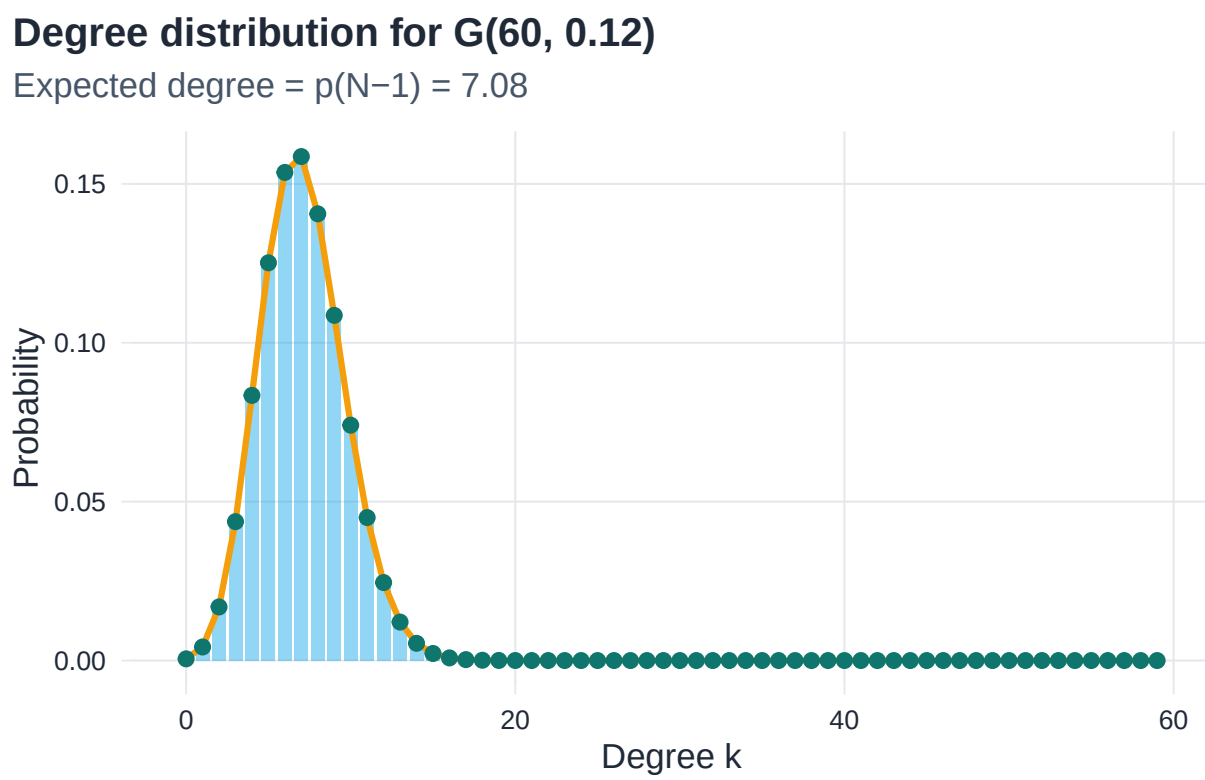


Figure 2.9: One realization of an Erdős–Rényi graph.

2.12 Interactive explorer

In the PDF version, the interactive widget is replaced by a static plot.



2.13 In-class questions

1. Why is a random graph a model rather than a single graph?
2. What is the main conceptual difference between $G(N, L)$ and $G(N, p)$?
3. Why does the total number of edges follow a binomial distribution in $G(N, p)$?
4. Why does the degree of a single node also follow a binomial distribution?
5. What does sparse mean in this chapter?
6. Why does the Poisson approximation appear in sparse large graphs?
7. Why is clustering in Erdős–Rényi graphs usually lower than in social networks?

2.14 End-of-chapter exercises

The exercises below are divided into three groups:

1. mathematical exercises,
2. conceptual and analytical exercises,
3. simulation and coding exercises.

They are designed to move from formal derivation to interpretation and numerical verification.

2.14.1 Part I. Mathematical exercises

2.14.1.1 Exercise 1. Counting graphs in $G(N, L)$

Let $G(N, L)$ be the Erdős–Rényi ensemble with N vertices and exactly L edges.

1. Show that the number of possible undirected edges is

$$K = \binom{N}{2}.$$

2. Prove that the number of labeled simple graphs in the ensemble is

$$\Omega(N, L) = \binom{K}{L}.$$

3. Deduce that the probability of any one graph in the ensemble is

$$P(G) = \frac{1}{\binom{K}{L}}.$$

2.14.1.2 Exercise 2. Counting graphs in $G(N, p)$

Let $G(N, p)$ be the Erdős–Rényi ensemble in which every possible edge is present independently with probability p.

1. Let a graph G contain exactly L edges. Show that its probability is

$$P(G) = p^L (1-p)^{K-L}, \qquad K = \binom{N}{2}.$$

2. Prove that all graphs with the same number of edges have the same probability.
3. Explain why graphs with different numbers of edges generally do not have the same probability.

2.14.1.3 Exercise 3. Edge probability in $G(N, L)$

In the ensemble $G(N, L)$, compute the probability that two fixed distinct vertices are connected by an edge.

Your derivation should start from combinatorial counting and end with a simplified formula in terms of N and L.

2.14.1.4 Exercise 4. Average degree in $G(N, L)$

Let $\bar{k}$ denote the average degree of a graph.

1. Prove that for every graph with N vertices and L edges,

$$\bar{k} = \frac{2L}{N}.$$

2. Deduce that in the ensemble $G(N, L)$,

$$\mathbb{E}[\bar{k}] = \frac{2L}{N}.$$

3. Explain why this quantity is not random in the ensemble $G(N, L)$.

2.14.1.5 Exercise 5. Average degree in $G(N,p)$

In the ensemble $G(N,p)$:

1. Show that the total number of edges satisfies
$$L \sim B\left(\binom{N}{2}, p\right).$$
2. Deduce that
$$\mathbb{E}[L] = p\binom{N}{2}.$$
3. Prove that
$$\mathbb{E}[\bar{k}] = p(N-1).$$

2.14.1.6 Exercise 6. Degree distribution in $G(N,p)$

Fix a vertex v in the model $G(N,p)$.

1. Show that the degree of v is a binomial random variable.
2. Derive the formula
$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}.$$
3. Compute its mean and variance.

2.14.1.7 Exercise 7. Poisson approximation

Assume that N is large, p is small, and

$$\lambda = p(N-1)$$

remains finite.

1. Starting from the exact degree distribution in $G(N,p)$,
$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k},$$
derive the Poisson approximation
$$P(k) \approx e^{-\lambda} \frac{\lambda^k}{k!}.$$
2. State clearly which approximations you use.
3. Explain why this approximation is natural in the sparse regime.

2.14.1.8 Exercise 8. Clustering coefficient in $G(N,p)$

Let i be a vertex of degree k_i.

1. Show that the maximum possible number of edges among the neighbors of i is
$$\binom{k_i}{2}.$$

2. Starting from the definition of local clustering coefficient, show that in $G(N, p)$,

$$\mathbb{E}[C_i] = p.$$

3. Deduce that

$$\mathbb{E}[\bar{C}] = p.$$

2.14.1.9 Exercise 9. Directed version of the models

Generalize the definitions of $G(N, L)$ and $G(N, p)$ to directed graphs without self-loops.

1. Determine the total number of possible directed edges.
2. Write the probability of a directed graph in the $G(N, L)$ version.
3. Write the probability of a directed graph in the $G(N, p)$ version.
4. Derive the expected in-degree and expected out-degree.

2.14.1.10 Exercise 10. Most probable graph size in $G(N, p)$

For fixed N and fixed p, the probability that a graph has exactly L edges is

$$P(L) = \binom{K}{L} p^L (1-p)^{K-L}, \qquad K = \binom{N}{2}.$$

Determine the value(s) of L that maximize this probability.

2.14.2 Part II. Conceptual and analytical exercises

2.14.2.1 Exercise 11. $G(N, L)$ versus $G(N, p)$

Write a short analytical comparison between the two models. Your answer should address:

- what is fixed in each model,
- what is random in each model,
- what it means for graphs to be equiprobable,
- why the two models are related but not identical.

2.14.2.2 Exercise 12. Same average degree, different structure

Construct two labeled graphs on the same number of vertices such that they have the same average degree but visibly different structure.

Discuss why average degree alone is insufficient to characterize a graph.

2.14.2.3 Exercise 13. Degree distribution versus average degree

Explain why knowing only $\mathbb{E}[\bar{k}]$ does not determine the degree distribution.

Your answer should include: - one statement about what average degree measures, - one statement about what the degree distribution measures, - one example showing that different degree distributions may have the same average degree.

2.14.2.4 Exercise 14. Why Erdős–Rényi clustering is small

Give a mathematical and conceptual explanation of why clustering in an Erdős–Rényi graph is usually low when p is small.

Your answer should distinguish between: - independent edge formation, - local triangle closure, - the behavior of social networks.

2.14.2.5 Exercise 15. Sparse versus dense regimes

Discuss the difference between the sparse and dense regimes in $G(N, p)$.

Your answer should address: - how p scales with N, - how $\mathbb{E}[k]$ behaves, - why the Poisson approximation belongs to the sparse regime.

2.14.2.6 Exercise 16. Why the Poisson law is only an approximation

The degree distribution in $G(N, p)$ is exactly binomial, but often approximated by a Poisson law.

Explain why the Poisson distribution is not exact in general, and state clearly the regime in which it becomes accurate.

2.14.3 Part III. Simulation and coding exercises

2.14.3.1 Exercise 17. Simulating $G(N, p)$ and estimating the degree distribution

Write an `R` function that:

1. generates m independent graphs from $G(N, p)$,
2. computes the empirical degree distribution across all simulated graphs,
3. compares it with the theoretical binomial distribution.

Use the function for several values of N and p.

2.14.3.2 Exercise 18. Binomial versus Poisson comparison

For a sequence of values of N and p such that $\lambda = p(N-1)$ remains approximately constant:

1. compute the exact binomial degree distribution,
2. compute the Poisson approximation,
3. plot both on the same axes,
4. discuss how the agreement changes as N increases.

2.14.3.3 Exercise 19. Comparing $G(N, L)$ and $G(N, p)$ numerically

Choose values of N, L, and p such that

$$L \approx p\binom{N}{2}.$$

Then:

1. generate one graph from $G(N, L)$,
2. generate one graph from $G(N, p)$,
3. compare their edge counts, average degree, and clustering,
4. repeat this many times and summarize the differences.

2.14.3.4 Exercise 20. Empirical verification of expected average degree

For several values of N and p, generate many graphs from $G(N, p)$ and compute the empirical mean of the average degree.

Then compare it with the theoretical value

$$p(N-1).$$

Discuss the effect of the number of repetitions on the accuracy of the estimate.

2.14.3.5 Exercise 21. Empirical verification of expected clustering

For several values of N and p, generate many graphs from $G(N, p)$ and compute the empirical mean of the average clustering coefficient.

Then compare it with the theoretical value

$$p.$$

Explain why small discrepancies may remain in finite simulations.

2.14.3.6 Exercise 22. Degree table and empirical distribution from one graph

Generate one graph from $G(N, p)$ and produce:

1. a table of vertex degrees,
2. a table of degree counts,
3. the empirical degree distribution,
4. a plot of the empirical distribution.

Then compare the empirical plot with the theoretical binomial curve.

2.14.3.7 Exercise 23. Sensitivity to parameter changes

Fix N and vary p over a grid of values.

For each value of p, compute:

- average degree,
- empirical clustering,
- largest observed degree.

Plot each quantity against p and discuss the trends.

2.14.3.8 Exercise 24. Simulation report

Prepare a short report of approximately two pages containing:

- one derivation from the theory part of the chapter,
- at least two simulation figures,
- one comparison of empirical and theoretical degree distributions,
- one discussion of average degree and clustering in $G(N, p)$.

2.15 Suggested mini-project

A strong mini-project for this chapter is the following:

> Compare the theoretical and empirical behavior of the Erdős–Rényi model under several parameter regimes.

A complete submission should include:

- mathematical derivations,
- simulation code,
- figures,
- interpretation of the results,
- a short concluding discussion.

2.16 Concluding remarks

The Erdős–Rényi model is not meant to reproduce every feature of real networks. Its value lies in its clarity.

It gives a first probabilistic language for talking about networks:

- a graph can be generated by a random rule
- many graph properties can be studied through expectations
- independent edge formation leads naturally to binomial laws
- sparse large graphs often produce Poisson-like degree distributions
- independent wiring tends to create low clustering

That is why Erdős–Rényi graphs remain such an important first benchmark in network science.

Chapter 3

The Evolution of Random Graphs

3.1 Introduction

The previous chapter focused on the static properties of random graphs: the Erdős–Rényi models, average degree, degree distribution, and clustering coefficient. Those quantities describe the graph at a fixed stage.

A second and equally important question is how a random graph changes as links are gradually introduced. At the beginning there may be no links at all, so every vertex is isolated. When a small number of links is added, tiny connected groups appear. As the density of links increases, these groups merge into larger connected structures. Eventually one component becomes comparable in size to the full graph. This object is called the giant component.

The central theme of this chapter is that the emergence of large-scale connectivity is governed by a threshold phenomenon. In the Erdős–Rényi model, the expected degree

$$\langle k \rangle = p(N-1)$$

is the key parameter controlling that transition.

3.2 Learning goals

By the end of this chapter, it should be possible to:

- define connected graphs and connected components precisely
- define the size of a component and the notion of a giant component
- explain how a random graph evolves as the edge probability increases
- distinguish the subcritical, critical, supercritical, and connected regimes
- explain why the threshold $\langle k \rangle = 1$ is important
- explain why the connectedness threshold occurs near $\langle k \rangle = \ln N$
- verify these ideas numerically in `R`

3.3 Connectedness and components

3.3.1 Connected graph

Definition 3.1. Let $G = (V, E)$ be an undirected graph. The graph G is called **connected** if for every pair of distinct vertices $u, v \in V$, there exists a path in G from u to v.

This definition means that the graph forms one reachable structure. Starting from any vertex, it is possible to move to any other vertex by following a sequence of existing edges.

Connectedness is therefore a global property. It is not determined by one degree value or by one local motif. It asks whether the graph, as a whole, holds together.

If this condition fails, then the graph is disconnected. In that case the graph splits into separate connected pieces.

3.3.2 Connected component

Definition 3.2. Let $G = (V, E)$ be an undirected graph. A subgraph H of G is called a **connected component** of G if:

1. H is connected, and
2. H is maximal with respect to inclusion among connected subgraphs of G.

The word *connected* means that every pair of vertices in H can be joined by a path entirely contained in H.

The word *maximal* is equally important. It means that H cannot be enlarged by adding another vertex of G while keeping the subgraph connected. So a connected component is not merely a connected part of the graph. It is a connected part that is already as large as possible.

This gives a natural decomposition of a disconnected graph: the graph is partitioned into its connected components.

Remark 3.1. The connected components of a graph form a partition of the vertex set V. Every vertex belongs to exactly one connected component.

3.3.3 Size of a component

Definition 3.3. Let H be a connected component of a graph $G = (V, E)$. The **size** of H is the number of vertices in H, namely

$$|V(H)|.$$

This quantity is simple, but it becomes crucial in random graph theory. As a random graph evolves, the central question is how the component-size pattern changes. Do only small components exist, or does one component become comparable to the full graph?

3.3.4 Isolated vertex

Definition 3.4. A vertex $v \in V$ is called **isolated** if its degree is zero.

An isolated vertex is a connected component of size 1. So isolated vertices are not exceptions to the theory of components. They are simply the smallest possible components.

3.3.5 Example: component structure

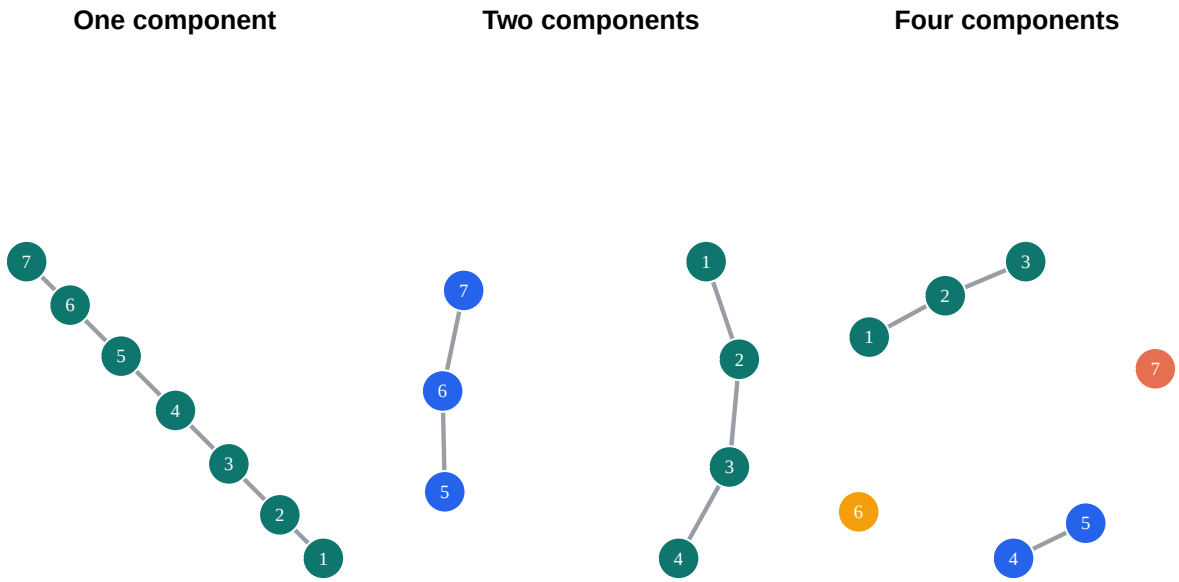


Figure 3.1: Three graphs with different component structure. Vertices in the same connected component share the same color.

The three graphs above illustrate the same idea in progressively more disconnected form:

- in the first graph, every vertex belongs to one single connected component
- in the second graph, the graph splits into two components
- in the third graph, the graph decomposes into several small pieces, including isolated vertices

3.3.6 Component-size table

Graph	Component sizes
One component	7
Two components	$4, 3$
Four components	$3, 2, 1, 1$

This table translates the picture into numerical structure and prepares the transition to the notion of a giant component.

3.4 Giant component

3.4.1 Definition

Definition 3.5. Let G_N be a graph on N vertices, and let $C_{\max}(G_N)$ denote its largest connected component. A sequence of random graphs is said to contain a **giant component** if

$$\frac{|C_{\max}(G_N)|}{N}$$

converges to a positive quantity as $N \to \infty$.

This definition should be interpreted carefully. Every finite graph has a largest connected component, but not every largest connected component is a giant component in the asymptotic sense.

The term *giant* means that the component contains a nonvanishing fraction of all vertices. It is therefore a macroscopic object, not merely the largest among several still-small pieces.

3.4.2 Why the giant component matters

The giant component marks the first stage at which the graph begins to behave like a true network at the global level.

Before it appears:

- most vertices belong only to tiny disconnected groups
- reachability is local
- no global connected backbone exists

After it appears:

- a positive fraction of vertices become mutually reachable
- one connected structure dominates the graph
- system-wide connectivity becomes possible

So the emergence of the giant component is the key event in the evolution of a random graph.

3.5 The evolution of a random graph

Consider the Erdős–Rényi model $G(N, p)$ with fixed N and variable parameter $p \in [0, 1]$.

At one extreme,

$$p = 0,$$

no edges are present, so every vertex is isolated.

At the other extreme,

$$p = 1,$$

every possible edge is present, so the graph is complete and hence connected.

The mathematically interesting behavior lies between these extremes. As p increases,

1. isolated vertices begin to join into tiny components,
2. small components merge into larger components,
3. one component begins to dominate,
4. eventually the graph becomes connected.

So the evolution of the graph is most naturally described through the evolution of its component structure.

3.5.1 Evolution through snapshots

To visualize this process clearly, the same vertex set and the same vertex positions are used in all snapshots below. The edge set evolves monotonically: once an edge appears, it remains present at later stages. This makes the sequence a true network evolution rather than a collection of unrelated random realizations.

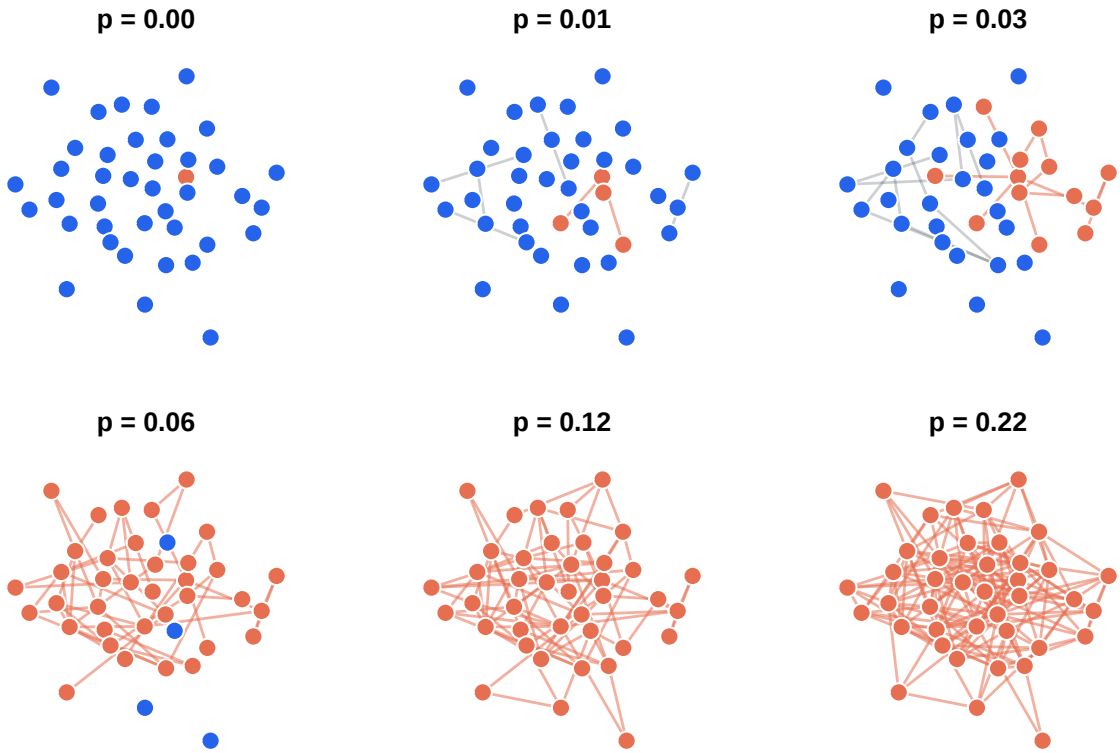


Figure 3.2: Evolution of a random graph on a fixed vertex set. Vertex positions are fixed across all snapshots, and the edge set grows monotonically with p. The largest connected component is highlighted.

These snapshots show several distinct structural stages.

- At $p = 0$, all vertices are isolated.
- For very small p, tiny components begin to appear.
- At intermediate values of p, components merge and one of them starts to dominate.
- At larger values of p, most vertices belong to a single large connected structure.

The qualitative transition is therefore from fragmentation to macroscopic connectivity.

3.6 Regimes of random-graph evolution

A convenient parameter for describing these regimes is the expected degree

$$\langle k \rangle = p(N-1).$$

This quantity is more informative than p alone because it measures the expected number of neighbors per vertex.

3.6.1 Subcritical regime

Definition 3.6. The random graph is in the **subcritical regime** when

$$\langle k \rangle < 1.$$

In this regime, no giant component exists. Almost all components are small, and isolated vertices are common. The graph contains connectivity, but only at a local scale.

3.6.2 Critical regime

Definition 3.7. The random graph is in the **critical regime** when

$$\langle k \rangle \approx 1.$$

This is the transition region. It is the stage at which the graph changes from a collection of small components into a graph containing a giant component.

In finite networks, the transition is not perfectly sharp. But as the system becomes large, the threshold becomes more pronounced.

3.6.3 Supercritical regime

Definition 3.8. The random graph is in the **supercritical regime** when

$$\langle k \rangle > 1.$$

This is the first regime in which a giant component exists. One connected component contains a positive fraction of all vertices, while many smaller components continue to coexist with it.

This regime is especially important because it is the first stage at which the graph has a genuine large-scale network structure.

3.6.4 Connected regime

Definition 3.9. The random graph enters the **connected regime** when the graph is connected with high probability as N becomes large. In the Erdős–Rényi model this occurs near

$$\langle k \rangle \approx \ln N,$$

or equivalently,

$$p \approx \frac{\ln N}{N}.$$

This threshold is much larger than the giant-component threshold $\langle k \rangle = 1$. So the emergence of a giant component and the graph becoming connected are two different events.

There is therefore a broad intermediate range in which the graph already has a giant component but is not yet fully connected.

3.7 Mathematical description of the giant component

A central quantity is the relative size of the giant component,

$$S = \frac{N_G}{N},$$

where N_G denotes the number of vertices in the largest connected component.

In the Erdős–Rényi model, the asymptotic fraction S of vertices contained in the giant component satisfies the self-consistency equation

$$S = 1 - e^{-\langle k \rangle S}.$$

This equation is a concise mathematical description of the giant component.

3.7.1 Interpretation of the self-consistency equation

The quantity $1 - S$ is the fraction of vertices that do **not** belong to the giant component.

A vertex fails to belong to the giant component only if all of its potential connections lead to vertices that also fail to belong to the giant component. In the Erdős–Rényi setting, that logic leads to the exponential form above.

So the equation

$$S = 1 - e^{-\langle k \rangle S}$$

expresses the self-consistent probability that a vertex belongs to the giant structure.

> **Key result: self-consistency equation**
>
> The fraction S of vertices in the giant component satisfies
>
> $$S = 1 - e^{-\langle k \rangle S}.$$
>
> This equation has no positive solution when $\langle k \rangle \leq 1$, and a unique positive solution when $\langle k \rangle > 1$.

3.7.2 Near-threshold behavior

When $\langle k \rangle$ is only slightly larger than 1, the giant component is still small relative to its later size, but it is already macroscopic. In that near-threshold region,

$$S \sim 2(\langle k \rangle - 1) \qquad \text{as } \langle k \rangle \downarrow 1^+.$$

Equivalently,

$$N_G \sim 2(\langle k \rangle - 1)N.$$

So near the transition, the giant component grows linearly with the distance from the threshold.

Near-threshold scaling

Just above the threshold, the giant component size grows as

$$S \approx 2(\langle k \rangle - 1), \qquad \langle k \rangle \downarrow 1^+.$$

The transition is therefore continuous: the giant component emerges gradually, growing linearly with the excess degree.

3.7.3 Structure in the supercritical regime

In the supercritical regime, the graph has two coexisting structural parts:

1. one giant component containing a positive fraction of all vertices,
2. many finite components outside the giant component.

Near the threshold, those finite components are typically small and predominantly tree-like. By contrast, the giant component contains many cycles and loops. So the onset of the giant component is also the onset of genuinely complex global connectivity.

3.7.4 Connectedness threshold

The giant component threshold occurs near

$$\langle k \rangle = 1,$$

but the graph becomes connected only much later, near

$$\langle k \rangle = \ln N.$$

This is a major conceptual point. The existence of a giant component does **not** imply that the graph is connected.

So there is an intermediate range,

$$1 < \langle k \rangle < \ln N,$$

in which

- the graph has a giant component,
- many smaller components still exist,
- isolated vertices may still be present,
- the full graph is not yet connected.

Only when the average degree reaches order $\ln N$ do these remaining disconnected pieces disappear with high probability.

Two distinct thresholds

The emergence of the giant component and the graph becoming fully connected are separate events:

- **Giant component threshold:** $\langle k \rangle = 1$
- **Connectedness threshold:** $\langle k \rangle \approx \ln N$

Between these two thresholds, the graph has a macroscopic connected backbone but also retains isolated vertices and small disconnected pieces.

3.8 Numerical exploration

The structural transition can be observed numerically by generating many graphs for different values of p and tracking several global quantities.

```
set.seed(2026)

n <- 500
k_grid <- seq(0.1, 8.0, length.out = 45)
p_grid <- k_grid / (n - 1)
reps <- 80

evolution_tbl <- map_dfr(p_grid, function(p) {
  map_dfr(seq_len(reps), function(rep_id) {
    g <- sample_gnp(n = n, p = p, directed = FALSE, loops = FALSE)
    comp <- components(g)$csize

    tibble(
      p = p,
      mean_degree = p * (n - 1),
      rep = rep_id,
      largest_component_fraction = max(comp) / n,
      number_of_components = length(comp),
      isolated_fraction = sum(comp == 1) / n,
      connected_indicator = as.integer(length(comp) == 1)
    )
  })
})

evolution_summary <- evolution_tbl |>
  group_by(p, mean_degree) |>
  summarise(
    largest_component_fraction = mean(largest_component_fraction),
    number_of_components = mean(number_of_components),
    isolated_fraction = mean(isolated_fraction),
    connected_probability = mean(connected_indicator),
    connected_probability_se = sqrt(connected_probability * (1 - connected_probability) / n()),
    .groups = "drop"
  )

iso_fit <- isoreg(evolution_summary$mean_degree, evolution_summary$connected_probability)
evolution_summary$connected_probability_iso <- iso_fit$yf
```

3.8.1 Largest component fraction

```
ggplot(evolution_summary, aes(x = mean_degree, y = largest_component_fraction)) +
  geom_line(color = course_cols$teal, linewidth = 1.2) +
  geom_point(color = course_cols$gold, size = 2.1) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Emergence of the giant component",
    subtitle = "Red dashed line: giant-component threshold. Blue dotted line: connectedness threshold."
    x = "Expected degree E[k]",
```

```
    y = "Largest component fraction"
  )
```

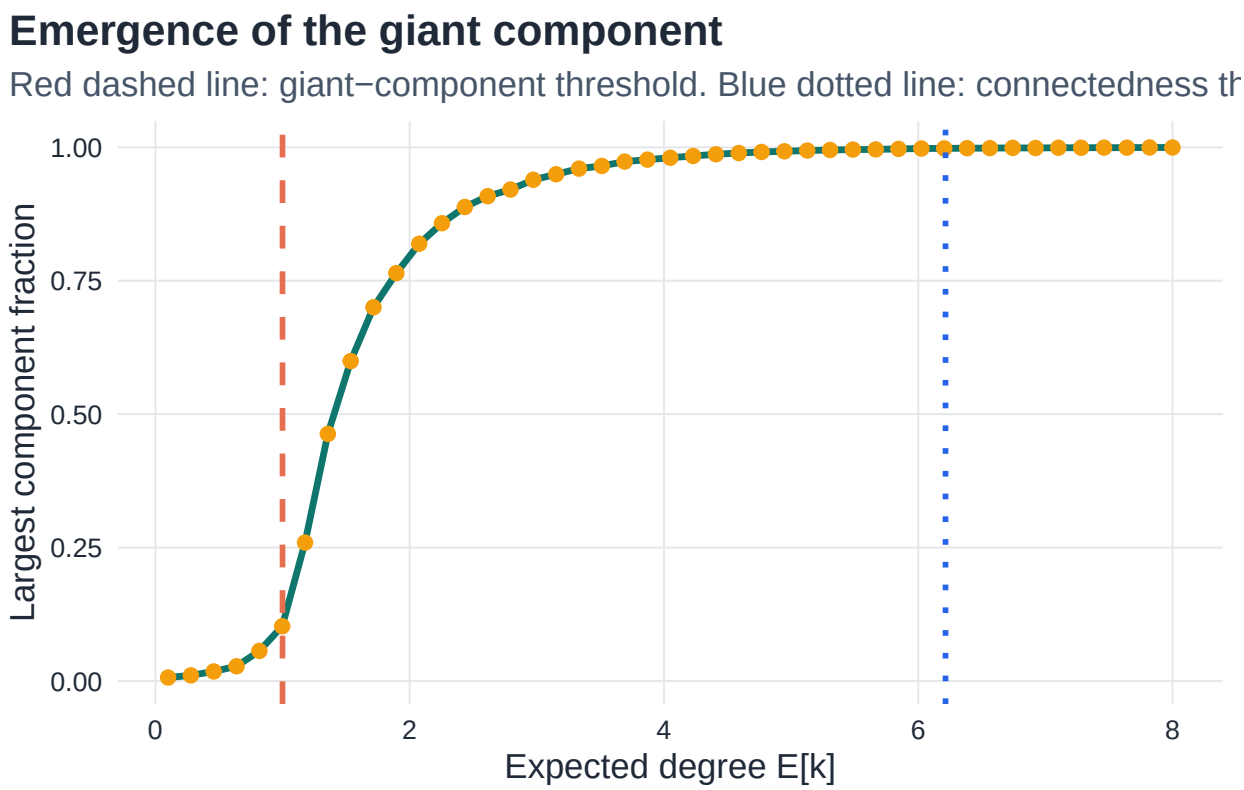


Figure 3.3: Average fraction of vertices in the largest connected component as a function of expected degree. The dashed vertical lines indicate the giant-component threshold and the connectedness threshold.

This plot captures the main phase transition. For small expected degree, the largest component occupies only a small fraction of the graph. Near $\langle k \rangle = 1$, the largest component begins to grow rapidly. Well before the graph becomes connected, one component already dominates the structure.

3.8.2 Number of connected components

```
ggplot(evolution_summary, aes(x = mean_degree, y = number_of_components)) +
  geom_line(color = course_cols$blue, linewidth = 1.2) +
  geom_point(color = course_cols$gold, size = 2.1) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Decrease in the number of connected components",
    subtitle = "Disconnected pieces merge as the graph evolves",
    x = "Expected degree E[k]",
    y = "Average number of components"
  )
```

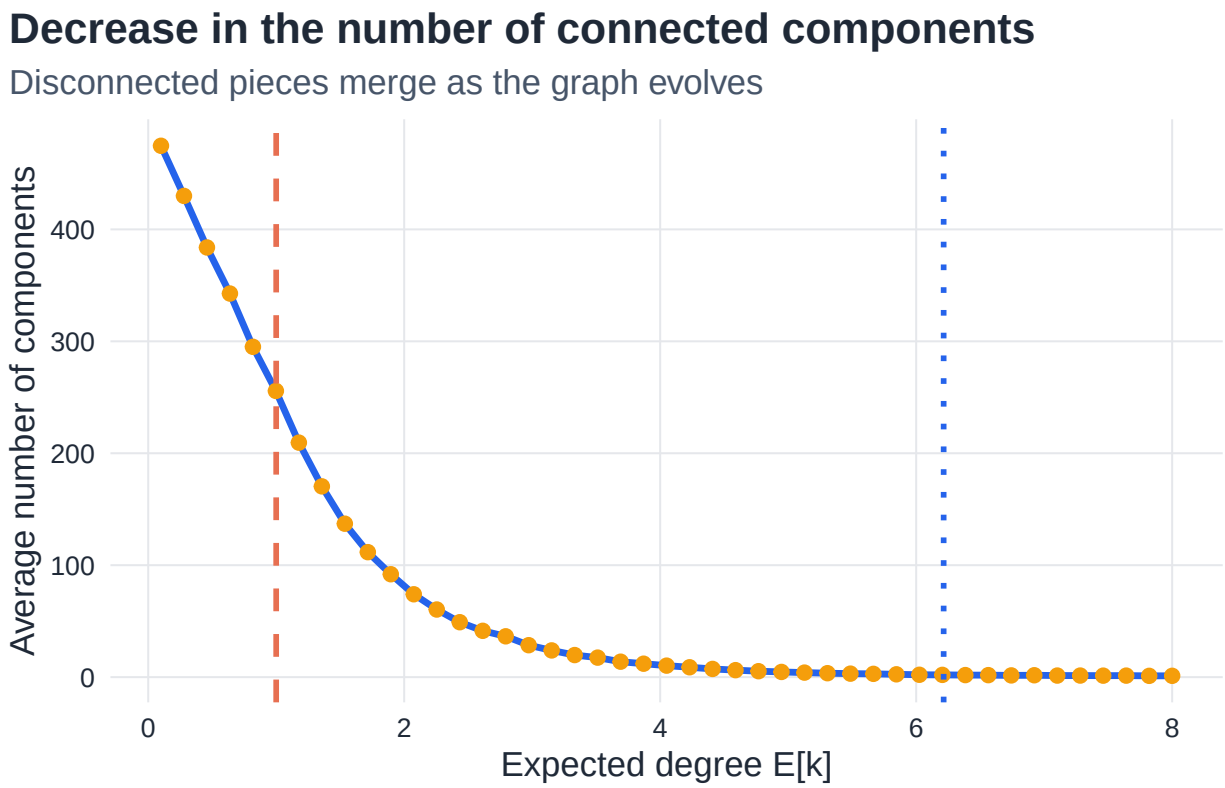


Figure 3.4: Average number of connected components as a function of expected degree.

This second plot reveals the same transition from another viewpoint. At small expected degree, the graph contains many disconnected pieces. As more links are added, these components merge, and the total number of components decreases.

3.8.3 Fraction of isolated vertices

```
ggplot(evolution_summary, aes(x = mean_degree, y = isolated_fraction)) +
  geom_line(color = course_cols$teal, linewidth = 1.2) +
  geom_point(color = course_cols$gold, size = 2.1) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Disappearance of isolated vertices",
    subtitle = "Isolated vertices become rare as the network densifies",
    x = "Expected degree E[k]",
    y = "Fraction of isolated vertices"
  )
```

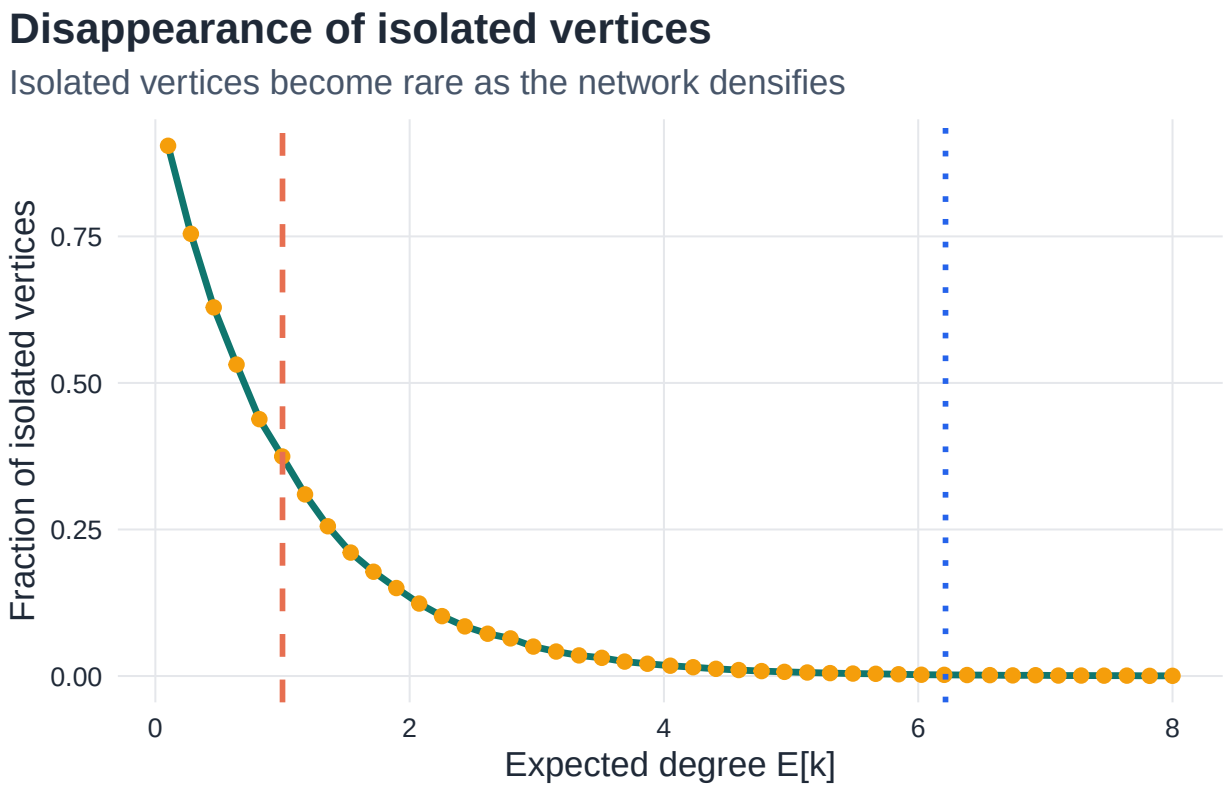


Figure 3.5: Average fraction of isolated vertices as a function of expected degree.

Isolated vertices are abundant at very small edge density, but they disappear as the graph evolves. Their disappearance is one of the main signals that the graph is approaching connectedness.

3.8.4 Probability of connectedness

```
ggplot(evolution_summary, aes(x = mean_degree)) +
  geom_ribbon(
    aes(
      ymin = pmax(connected_probability - 1.96 * connected_probability_se, 0),
      ymax = pmin(connected_probability + 1.96 * connected_probability_se, 1)
    ),
    fill = course_cols$blue,
    alpha = 0.12
  ) +
  geom_point(aes(y = connected_probability), color = course_cols$gold, size = 1.9) +
  geom_line(aes(y = connected_probability_iso), color = course_cols$coral, linewidth = 1.2) +
  geom_vline(xintercept = 1, color = course_cols$coral, linewidth = 1.0, linetype = 2) +
  geom_vline(xintercept = log(n), color = course_cols$blue, linewidth = 1.0, linetype = 3) +
  labs(
    title = "Connectedness probability",
    subtitle = "The true probability is monotone increasing; the fitted line enforces that theoretical p
    x = "Expected degree E[k]",
    y = "Probability that the graph is connected"
  )
```

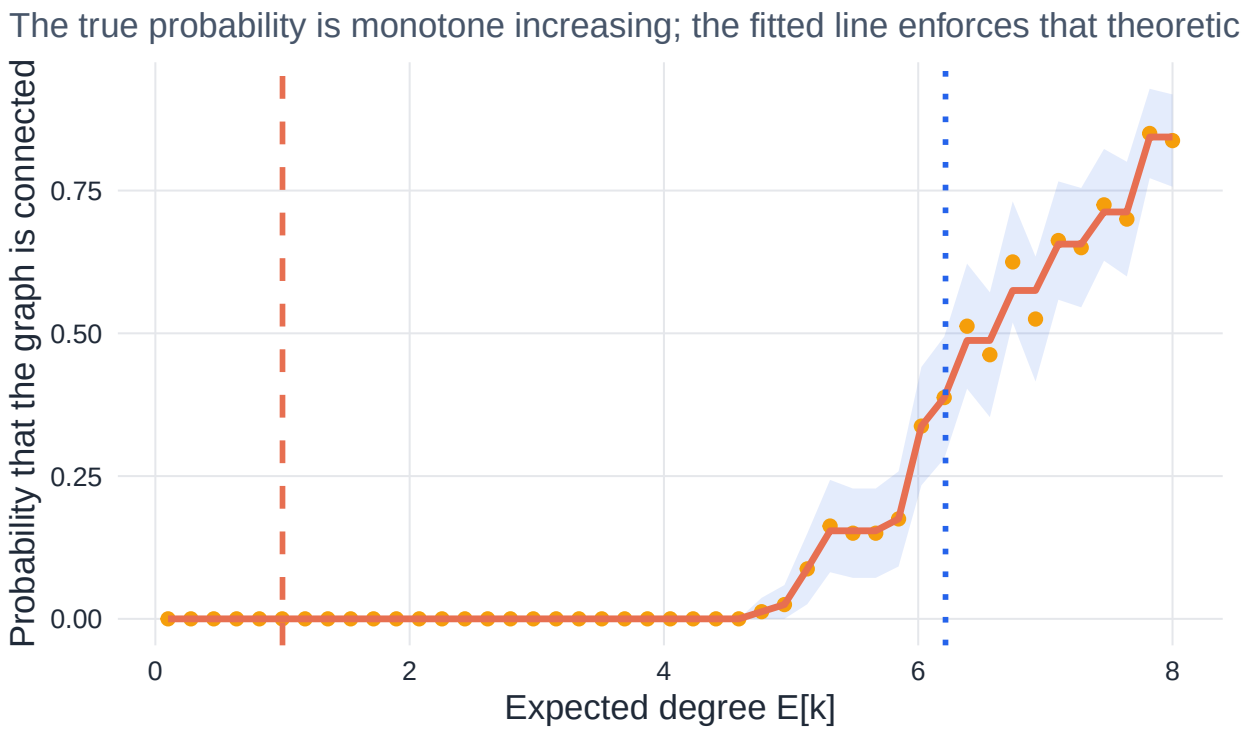


Figure 3.6: Estimated connectedness probability as a function of expected degree. Points show Monte Carlo estimates; the solid curve is the monotone isotonic fit.

For fixed N, the true connectedness probability is a monotone increasing function of p. Small downward wiggles in raw simulation estimates are caused by Monte Carlo noise. The points in the figure are the empirical estimates from a finite number of repetitions, while the solid line is an isotonic fit, which enforces the correct monotone behavior.

This last plot makes the distinction between the two thresholds especially clear. The giant component appears near $\langle k \rangle = 1$, but the probability that the graph is fully connected remains small until the expected degree is of order $\ln N$.

3.9 Summary of regimes

Regime	Condition	Main structural property
Subcritical	$\langle k \rangle < 1$	Only small connected components are present
Critical	$\langle k \rangle \approx 1$	The giant component begins to emerge
Supercritical	$\langle k \rangle > 1$	A giant component coexists with many finite components
Connected	$\langle k \rangle \approx \ln N$	The full graph is connected with high probability

This table summarizes the evolution of the Erdős–Rényi random graph in terms of its component structure.

3.10 Core formulas to remember

Summary

For the evolution of Erdős–Rényi random graphs $G(N, p)$:

1. **Expected degree:**
$$\langle k \rangle = p(N-1)$$
2. **Giant-component threshold:**
$$\langle k \rangle = 1 \quad \Leftrightarrow \quad p_c \approx \frac{1}{N}$$
3. **Giant-component fraction** (self-consistency equation):
$$S = 1 - e^{-\langle k \rangle S}$$
4. **Near-threshold size:**
$$S \approx 2(\langle k \rangle - 1), \qquad \langle k \rangle \downarrow 1^+$$
5. **Connectedness threshold:**
$$\langle k \rangle \approx \ln N \quad \Leftrightarrow \quad p \approx \frac{\ln N}{N}$$

3.11 In-class questions

1. Why is connectedness a global property of a graph?
2. What role does maximality play in the definition of a connected component?
3. Why is an isolated vertex still a connected component?
4. What distinguishes a largest component from a giant component?
5. Why does the threshold $\langle k \rangle = 1$ matter?
6. Why does the connectedness threshold occur later, near $\langle k \rangle = \ln N$?
7. Why can a graph have a giant component and still be disconnected?

3.12 End-of-chapter exercises

The exercises below are grouped into three categories:

1. mathematical exercises,
2. conceptual and analytical exercises,
3. simulation and coding exercises.

They are designed to move from formal understanding to computational verification.

3.12.1 Part I. Mathematical exercises

3.12.1.1 Exercise 1. Components and partitions

Let $G = (V, E)$ be an undirected graph.

1. Prove that every vertex of G belongs to at least one connected component.
2. Prove that no vertex can belong to two distinct connected components.
3. Deduce that the connected components of G form a partition of the vertex set V.

3.12.1.2 Exercise 2. Isolated vertices

Let $v \in V$ be a vertex with degree zero.

1. Prove that $\{v\}$ is a connected subgraph.
2. Prove that $\{v\}$ is maximal among connected subgraphs containing v.
3. Conclude that an isolated vertex is a connected component of size 1.

3.12.1.3 Exercise 3. Giant component criterion

Let $C_{\max}(G_N)$ denote the largest connected component of a graph on N vertices.

1. Explain why every finite graph has a largest connected component.
2. Give an example of a sequence of graphs for which
$$\frac{|C_{\max}(G_N)|}{N} \to 0.$$
3. Give an example of a sequence of graphs for which
$$\frac{|C_{\max}(G_N)|}{N} \to c > 0.$$
4. Explain why the second case corresponds to the existence of a giant component.

3.12.1.4 Exercise 4. Threshold written in terms of p

In the Erdős–Rényi model $G(N, p)$, the expected degree is

$$\langle k \rangle = p(N-1).$$

1. Show that the condition
$$\langle k \rangle = 1$$
implies
$$p_c = \frac{1}{N-1}.$$
2. Show that for large N this becomes
$$p_c \sim \frac{1}{N}.$$
3. Explain why this implies that the critical edge density becomes very small in large networks.

3.12.1.5 Exercise 5. Connectedness threshold

The connectedness threshold in the Erdős–Rényi model occurs near

$$\langle k \rangle \approx \ln N.$$

1. Show that this corresponds to
$$p \approx \frac{\ln N}{N-1}.$$
2. Compare this threshold with the giant-component threshold
$$p_c \approx \frac{1}{N-1}.$$

3. Prove that

$$\ln N > 1 \qquad \text{for all } N > e,$$

and explain why connectedness must occur later than the emergence of the giant component.

3.12.1.6 Exercise 6. The self-consistency equation

The asymptotic fraction S of nodes in the giant component satisfies

$$S = 1 - e^{-\langle k \rangle S}.$$

1. Verify that $S = 0$ is always a solution.
2. Show that if $\langle k \rangle < 1$, then the tangent of the right-hand side at $S = 0$ has slope smaller than 1.
3. Explain why this indicates that no positive solution exists in the subcritical regime.
4. Show that when $\langle k \rangle > 1$, the slope at $S = 0$ exceeds 1.
5. Explain why this allows the existence of a positive solution corresponding to a giant component.

3.12.1.7 Exercise 7. Near-threshold expansion

Starting from

$$S = 1 - e^{-\langle k \rangle S},$$

assume that S is small and that $\langle k \rangle = 1 + \varepsilon$ with $\varepsilon > 0$ small.

1. Expand the exponential up to second order in S.
2. Show that, to leading order,

$$S \approx 2(\langle k \rangle - 1).$$

3. Interpret this result in terms of the growth of the giant component just above the threshold.

3.12.2 Part II. Conceptual and analytical exercises

3.12.2.1 Exercise 8. Regime classification

For each of the following values of $\langle k \rangle$, determine whether the graph is expected to be in the subcritical, critical, supercritical, or connected regime:

1. $\langle k \rangle = 0.4$
2. $\langle k \rangle = 0.95$
3. $\langle k \rangle = 1.8$
4. $\langle k \rangle = \ln 500$

In each case, describe the expected component structure.

3.12.2.2 Exercise 9. Giant component versus connected graph

Explain carefully why the following two statements are not equivalent:

1. the graph has a giant component,
2. the graph is connected.

Your answer should include:

- a formal distinction,
- an explanation in terms of component sizes,
- an example of a graph that has a giant component but is not connected.

3.12.2.3 Exercise 10. Structural interpretation of the regimes

Write a short mathematical discussion comparing the four regimes:

- subcritical,
- critical,
- supercritical,
- connected.

For each regime, describe:

- the typical component sizes,
- whether isolated vertices are common,
- whether a giant component exists,
- whether the graph is connected.

3.12.2.4 Exercise 11. Why the emergence is a phase transition

The emergence of the giant component is often described as a phase transition.

Write a short explanation of this statement. Your answer should address:

- what quantity changes qualitatively,
- what the threshold parameter is,
- why the transition is not merely a gradual visual change,
- why this phenomenon is global rather than local.

3.12.2.5 Exercise 12. Trees and cycles outside the giant component

In the supercritical regime, many finite components outside the giant component are tree-like.

1. Explain why a finite tree on m vertices has exactly $m - 1$ edges.
2. Explain why the appearance of cycles indicates additional structural complexity.
3. Discuss why the giant component is expected to contain cycles, whereas many small finite components remain tree-like near the threshold.

3.12.3 Part III. Simulation and coding exercises

3.12.3.1 Exercise 13. Component-size computation

Write an `R` function that takes an undirected graph `g` and returns:

- the number of connected components,
- the vector of component sizes,
- the size of the largest connected component,
- the fraction of vertices in the largest connected component.

Test the function on at least three hand-constructed graphs.

3.12.3.2 Exercise 14. Evolution of the largest component

For fixed $N = 500$, simulate the model $G(N, p)$ over a grid of p values corresponding to

$$0.1 \leq \langle k \rangle \leq 8.$$

For each value of p, estimate the average largest-component fraction over repeated realizations.

Tasks:

1. plot the largest-component fraction against $\langle k \rangle$,
2. indicate the line $\langle k \rangle = 1$,
3. explain where the giant component begins to emerge.

3.12.3.3 Exercise 15. Number of components and isolated vertices

Using the same simulations as in Exercise 14, compute and plot:

1. the average number of connected components,
2. the average fraction of isolated vertices.

Discuss how these two quantities evolve as $\langle k \rangle$ increases.

3.12.3.4 Exercise 16. Empirical connectedness probability

For fixed $N = 500$, estimate the probability that $G(N, p)$ is connected for a grid of values of p.

Tasks:

1. plot the empirical connectedness probability against $\langle k \rangle$,
2. indicate the line $\langle k \rangle = \ln N$,
3. explain why the curve should be monotone increasing in theory,
4. explain why Monte Carlo estimates may show small fluctuations.

3.12.3.5 Exercise 17. Finite-size comparison

Repeat the largest-component simulation for

$$N = 100, \quad 250, \quad 500, \quad 1000.$$

For each value of N:

1. plot the largest-component fraction against $\langle k \rangle$,
2. compare the sharpness of the transition near $\langle k \rangle = 1$,
3. discuss how the finite-size behavior changes as N increases.

3.12.3.6 Exercise 18. Connectedness threshold and system size

For each of the values

$$N = 100, \quad 250, \quad 500, \quad 1000,$$

estimate numerically the value of $\langle k \rangle$ at which the graph becomes connected with probability approximately 0.5.

Then compare your estimates with $\ln N$.

Discuss whether the numerical behavior supports the theoretical scaling of the connectedness threshold.

3.12.3.7 Exercise 19. Monotone network evolution

Construct a sequence of graphs on a fixed set of vertices such that the edge set grows monotonically as p increases.

Tasks:

1. keep vertex positions fixed across all snapshots,
2. display at least six stages of evolution,
3. highlight the largest component at each stage,
4. explain how the component structure changes visually.

3.12.3.8 Exercise 20. Numerical solution of the giant-component equation

Consider the equation

$$S = 1 - e^{-\langle k \rangle S}.$$

1. Write an `R` function that solves this equation numerically for a given value of $\langle k \rangle$.
2. Compute the solution for a grid of values of $\langle k \rangle$ between 0 and 5.
3. Plot the resulting theoretical curve.
4. Compare it with your simulation-based estimate of the largest-component fraction.

3.12.3.9 Exercise 21. Short computational report

Prepare a short report of approximately two pages based on your simulations. The report should include:

- a brief statement of the model,
- at least two figures,
- one discussion of the threshold near $\langle k \rangle = 1$,
- one discussion of the connectedness threshold near $\langle k \rangle = \ln N$,
- a concluding paragraph interpreting the emergence of global connectivity.

3.13 Suggested project-style extension

A strong mini-project for this chapter is the following:

> Compare the simulated evolution of $G(N, p)$ with the theoretical equations for giant-component emergence and connectedness threshold.

A complete submission should contain:

- mathematical derivations,
- simulation code,
- plots,
- interpretation of the results.

3.14 Concluding remarks

The evolution of a random graph is one of the clearest demonstrations of how global structure can arise from simple local randomness.

At first the graph contains only isolated vertices and very small components. As edges are added, these small pieces merge. Near the threshold $\langle k \rangle = 1$, a giant component emerges. Much later, near $\langle k \rangle = \ln N$, the graph becomes connected with high probability.

So the emergence of a network is not a smooth undifferentiated process. It is organized by precise structural thresholds, and those thresholds can be described mathematically in terms of component size and expected degree.

Chapter 4

Small-World Networks

4.1 Introduction

The small-world phenomenon is one of the most widely discussed ideas in network science. In popular language it often appears through the phrase *six degrees of separation*: two individuals, even if they live very far apart, may still be connected through only a short chain of acquaintances.

From the viewpoint of graph theory, however, the important issue is not whether the number is exactly six. The real content of the statement is structural. It says that the distance between two randomly chosen vertices in a large network is often surprisingly small.

This immediately raises two mathematical questions:

1. What does *small* mean?
2. Relative to what benchmark is this smallness surprising?

The first question leads to the concepts of distance, diameter, and average path length. The second leads to a comparison between random graphs and regular lattices. A random graph typically has distances that grow only logarithmically with system size, whereas regular lattices exhibit polynomial growth. That contrast is the mathematical source of the small-world effect.

A further question then appears. Random graphs can have short distances, but they usually have low clustering. Many real systems, by contrast, display both:

- short path lengths,
- high clustering.

The final goal of the chapter is therefore to understand how these two properties can coexist. The Watts–Strogatz model provides one of the first and most influential answers to that question.

4.2 Learning goals

By the end of this chapter, it should be possible to:

- define graph distance, diameter, and average path length precisely
- explain the small-world property in mathematical terms
- derive the logarithmic distance estimate for random networks
- explain why small worlds are surprising when compared with regular lattices
- define the Watts–Strogatz model
- explain how shortcuts reduce path length while preserving clustering
- identify the small-world regime numerically in `R`
- discuss why small-world structure is relevant in applications

4.3 Distance, diameter, and average path length

4.3.1 Distance

Definition 4.1. Let $G = (V, E)$ be a connected undirected graph. For vertices $u, v \in V$, the **distance** between u and v, denoted by

$$d(u, v),$$

is the length of a shortest path from u to v.

This is the fundamental metric quantity in a graph. It counts the minimum number of edges that must be traversed in order to move from u to v.

If u and v are adjacent, then $d(u, v) = 1$. If they are not adjacent but share a common neighbor, then $d(u, v) = 2$, and so on.

4.3.2 Diameter

Definition 4.2. Let $G = (V, E)$ be a connected graph. The **diameter** of G is

$$D(G) = \max_{u,v \in V} d(u, v).$$

The diameter measures the largest distance between any pair of vertices in the graph. It is therefore a metric measure of the size of the network.

Because it depends on the most extreme pair of vertices, the diameter is often sensitive to rare structural features.

4.3.3 Average path length

Definition 4.3. Let $G = (V, E)$ be a connected graph with $N = |V|$ vertices. The **average path length** of G is

$$L(G) = \frac{1}{N(N-1)} \sum_{\substack{u,v \in V \\ u \neq v}} d(u, v).$$

This quantity measures the typical separation between two vertices in the graph.

It is usually more stable than the diameter, because it averages distances over all ordered pairs. In teaching and in applications, it is often the more informative quantity, since it describes typical rather than extreme reachability.

4.3.4 Why averaging suppresses fluctuations

The distinction between diameter and average path length is important.

- The diameter is controlled by a few extreme paths.
- The average path length is computed over all vertex pairs.

As a consequence, the averaging process suppresses fluctuations. This is one reason why logarithmic distance estimates are often closer to the average path length than to the diameter.

4.3.5 Clustering coefficient

Definition 4.4. Let $G = (V, E)$ be an undirected graph. The **average clustering coefficient** of G is

$$\bar{C}(G) = \frac{1}{N} \sum_{i=1}^{N} C_i,$$

where C_i denotes the local clustering coefficient of vertex i.

Clustering is a local quantity. It measures how strongly neighborhoods close into triangles.

The central question of this chapter is whether a network can have simultaneously:

- small global distances,
- strong local clustering.

That coexistence is the essence of the small-world property.

4.4 The small-world property in random networks

A first explanation of short distances comes from a simple random-network calculation.

4.4.1 A branching approximation

Suppose a network has average degree $\langle k \rangle$, and suppose its local neighborhoods are approximately tree-like.

Start from one vertex. Then, at a crude homogeneous level:

- there are about $\langle k \rangle$ vertices at distance 1,
- about $\langle k \rangle^2$ vertices at distance 2,
- about $\langle k \rangle^3$ vertices at distance 3,
- and more generally about $\langle k \rangle^d$ vertices at distance d.

This approximation ignores overlap between neighborhoods and therefore is not exact. But it captures the main mechanism: the number of reachable vertices grows exponentially with distance.

A more refined branching-process argument would replace $\langle k \rangle$ by an effective branching factor. The central conclusion, however, is the same: exponential neighborhood growth leads to logarithmic distances.

4.4.2 Number of vertices within distance d

Under the same approximation, the number of vertices reachable within distance at most d is

$$N(d) \approx 1 + \langle k \rangle + \langle k \rangle^2 + \cdots + \langle k \rangle^d.$$

This is a finite geometric series, so

$$N(d) \approx \frac{\langle k \rangle^{d+1} - 1}{\langle k \rangle - 1}.$$

The significance of this formula is that reachability expands very quickly when $\langle k \rangle > 1$.

4.4.3 Logarithmic estimate for distances

The quantity $N(d)$ cannot exceed the total number of vertices N. So the largest relevant distance scale must satisfy approximately

$$N(d_{\max}) \approx N.$$

If $\langle k \rangle$ is larger than one and N is large, the dominant term in the geometric sum is the last one, giving the simplified relation

$$\langle k \rangle^{d_{\max}} \approx N.$$

Taking logarithms yields

$$d_{\max} \approx \frac{\ln N}{\ln \langle k \rangle}.$$

This is the first mathematical form of the small-world phenomenon.

4.4.4 Average path length in random networks

In many networks, the expression

$$\frac{\ln N}{\ln \langle k \rangle}$$

gives a better approximation to the average path length than to the diameter. Hence one often writes

$$L(G) \approx \frac{\ln N}{\ln \langle k \rangle}.$$

This formula has a clear interpretation.

1. The dependence on N is logarithmic.
2. The dependence on degree is through the factor $1/\ln \langle k \rangle$.

Since $\ln N \ll N$, the logarithmic law means that distances grow extremely slowly with system size. This is what *small* means in the expression *small world*: the distance scale is proportional to $\ln N$, not to N or to a power of N.

Key result: logarithmic distance scaling

In a random network with N vertices and average degree $\langle k \rangle > 1$, the average path length satisfies

$$L(G) \approx \frac{\ln N}{\ln \langle k \rangle}.$$

Distances grow only logarithmically with system size — far slower than the polynomial growth $N^{1/d}$ seen in d-dimensional regular lattices.

4.4.5 A social-network illustration

Take approximate values

$$N \approx 7 \times 10^9, \qquad \langle k \rangle \approx 10^3.$$

Then

$$L(G) \approx \frac{\ln(7 \times 10^9)}{\ln(10^3)} \approx 3.28.$$

This rough estimate suggests that the typical social separation scale is only a few steps. The exact numerical value should not be taken literally, but the order of magnitude is the key message: logarithmic scaling leads to surprisingly short paths even in extremely large systems.

4.5 Why small worlds are surprising

The small-world effect is surprising because our geometric intuition is often shaped by regular lattices, and regular lattices do **not** display logarithmic distance growth.

4.5.1 Distance scaling in regular lattices

In a one-dimensional lattice,

$$L(G) \sim D(G) \sim N.$$

In a two-dimensional square lattice,

$$L(G) \sim D(G) \sim N^{1/2}.$$

In a three-dimensional cubic lattice,

$$L(G) \sim D(G) \sim N^{1/3}.$$

More generally, in a d-dimensional regular lattice,

$$L(G) \sim D(G) \sim N^{1/d}.$$

These are polynomial growth laws, not logarithmic ones.

4.5.2 Why the contrast matters

This contrast is mathematically significant. Polynomial growth is much faster than logarithmic growth when N is large.

For example, if a population-scale network were arranged like a two-dimensional lattice, then the characteristic separation scale would be of order $N^{1/2}$, which is vastly larger than the logarithmic random-network estimate.

So the small-world effect is not simply the fact that distances are finite. It is the much stronger fact that they are drastically smaller than what regular lattice geometry would suggest.

4.5.3 Simulated scaling of average path length

The purpose of this subsection is to compare the *growth laws* of average path length across several network families. For this reason, the values of N do not need to be identical across all models. What matters is that each family is sampled on a sufficiently fine size grid so that its scaling trend is visible.

This is especially important for lattice models. In three dimensions, for example, the natural graph sizes are cubes, so forcing one common set of N values creates large gaps and produces visually poor scaling plots. It is therefore more informative to let each family use its own sequence of system sizes.

The observable is the average path length rather than the diameter. This is a deliberate methodological choice. The diameter is determined by a few extreme vertex pairs and is therefore more irregular, while the average path length is computed over all vertex pairs, so the averaging process suppresses fluctuations and provides a much more stable quantity for scaling analysis.

The simulated models are:

- a one-dimensional lattice,
- a two-dimensional lattice,
- a three-dimensional lattice,
- a connected Erdős–Rényi random graph with expected degree approximately equal to 8.

```r
library(tidyverse)
library(igraph)

mean_apl <- function(g) {
  mean_distance(g, directed = FALSE, unconnected = FALSE)
}

simulate_connected_er_apl <- function(n, mean_degree = 8, reps = 20) {
  p <- mean_degree / (n - 1)

  vals <- replicate(reps, {
    repeat {
      g <- sample_gnp(n, p, directed = FALSE, loops = FALSE)
      if (components(g)$no == 1) break
    }
    mean_apl(g)
  })

  mean(vals)
}

apl_lattice <- function(...) {
  g <- make_lattice(dimvector = c(...), periodic = FALSE)
  mean_apl(g)
}

line_sizes   <- c(20, 30, 40, 50, 60, 80, 100)
cube_sides   <- c(3, 4, 5, 6, 7, 8)
random_sizes <- c(20, 30, 40, 50, 60, 80, 100, 120, 150, 180, 220, 260, 320, 400)

rect_dims <- tribble(
  ~a, ~b,
   4,  4,
   5,  5,
   6,  6,
```

```r
   7,  7,
   8,  8,
   9,  9,
  10, 10,
  10, 12,
  12, 12,
  12, 15,
  15, 15,
  15, 18,
  18, 18,
  20, 20
)

distance_scaling_tbl <- bind_rows(
  tibble(
    N = line_sizes,
    avg_distance = map_dbl(line_sizes, ~ apl_lattice(.x)),
    model = "1D lattice"
  ),
  tibble(
    N = rect_dims$a * rect_dims$b,
    avg_distance = map2_dbl(rect_dims$a, rect_dims$b, apl_lattice),
    model = "2D lattice"
  ),
  tibble(
    N = cube_sides^3,
    avg_distance = map_dbl(cube_sides, ~ apl_lattice(.x, .x, .x)),
    model = "3D lattice"
  ),
  tibble(
    N = random_sizes,
    avg_distance = map_dbl(random_sizes, ~ simulate_connected_er_apl(.x, mean_degree = 8, reps = 8)),
    model = "Random graph"
  )
) |>
  mutate(
    model = factor(model, levels = c("1D lattice", "2D lattice", "3D lattice", "Random graph")),
    logN = log(N),
    logd = log(avg_distance)
  )
```

```r
ggplot(distance_scaling_tbl, aes(x = N, y = avg_distance, color = model)) +
  geom_line(linewidth = 1.15) +
  geom_point(size = 2.4) +
  scale_color_manual(
    values = c(
      "1D lattice"   = course_cols$blue,
      "2D lattice"   = course_cols$teal,
      "3D lattice"   = course_cols$gold,
      "Random graph" = course_cols$slate
    )
  ) +
  labs(
    title = "Average path length versus system size",
```

```
    subtitle = "Scaling behavior across lattice dimensions and random graphs",
    x = "Number of vertices (N)",
    y = "Average path length",
    color = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 15),
    plot.subtitle = element_text(size = 11, color = "grey30"),
    axis.title = element_text(face = "bold"),
    legend.position = "top",
    panel.grid.minor = element_blank()
  )
```

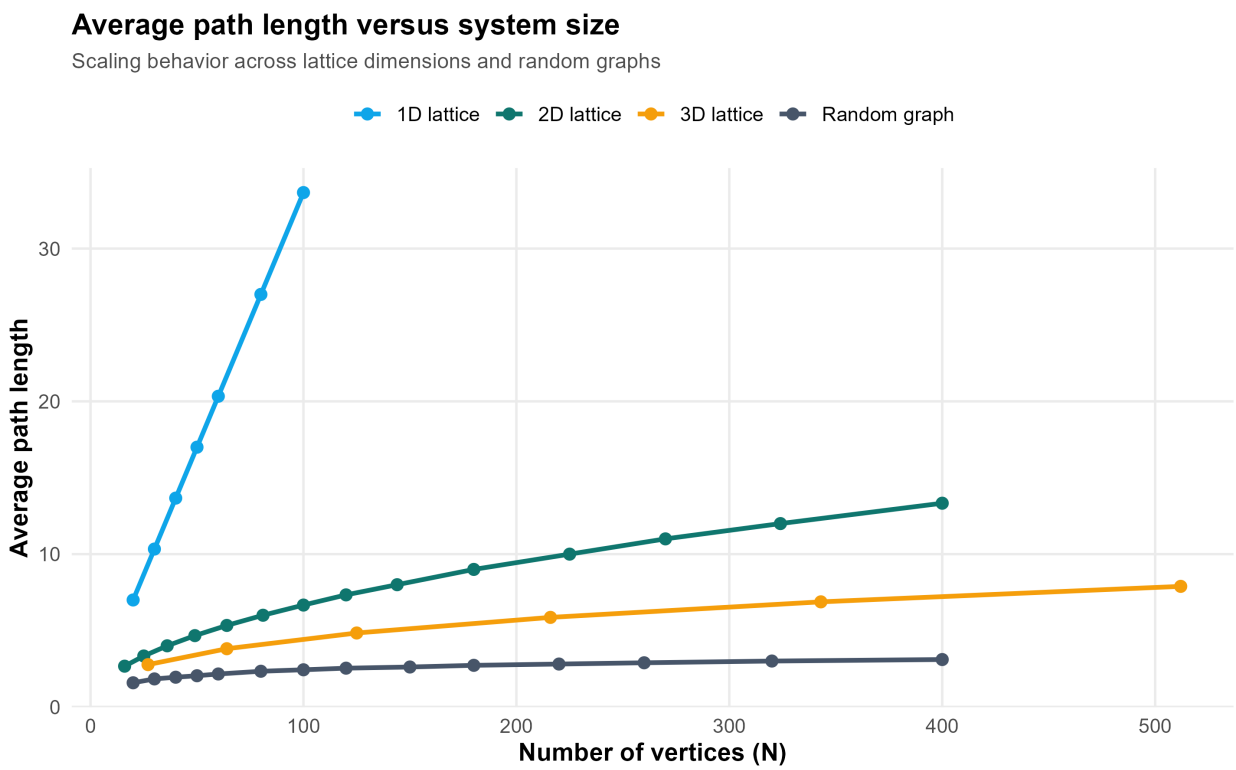


Figure 4.1: Average path length as a function of system size for 1D, 2D, 3D lattices, and random graphs.

The linear-scale plot already shows the main contrast. The one-dimensional lattice grows fastest, the two-dimensional and three-dimensional lattices grow more slowly, and the random graph grows much more slowly still.

To interpret the scaling laws more clearly, it is useful to look at the logarithms directly.

```
ggplot(distance_scaling_tbl, aes(x = logN, y = logd, color = model)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2.1) +
  scale_color_manual(
    values = c(
      "1D lattice" = course_cols$blue,
      "2D lattice" = course_cols$teal,
```

```
    "3D lattice" = course_cols$gold,
    "Random graph" = course_cols$slate
  )
) +
labs(
  title = "Logarithmic view of distance scaling",
  subtitle = "Lattice growth is polynomial, while random-network growth is much slower",
  x = "ln N",
  y = "ln <d>",
  color = NULL
) +
theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(face = "bold", size = 15),
  plot.subtitle = element_text(size = 11, color = "grey30"),
  axis.title = element_text(face = "bold"),
  legend.position = "top",
  panel.grid.minor = element_blank()
)
```

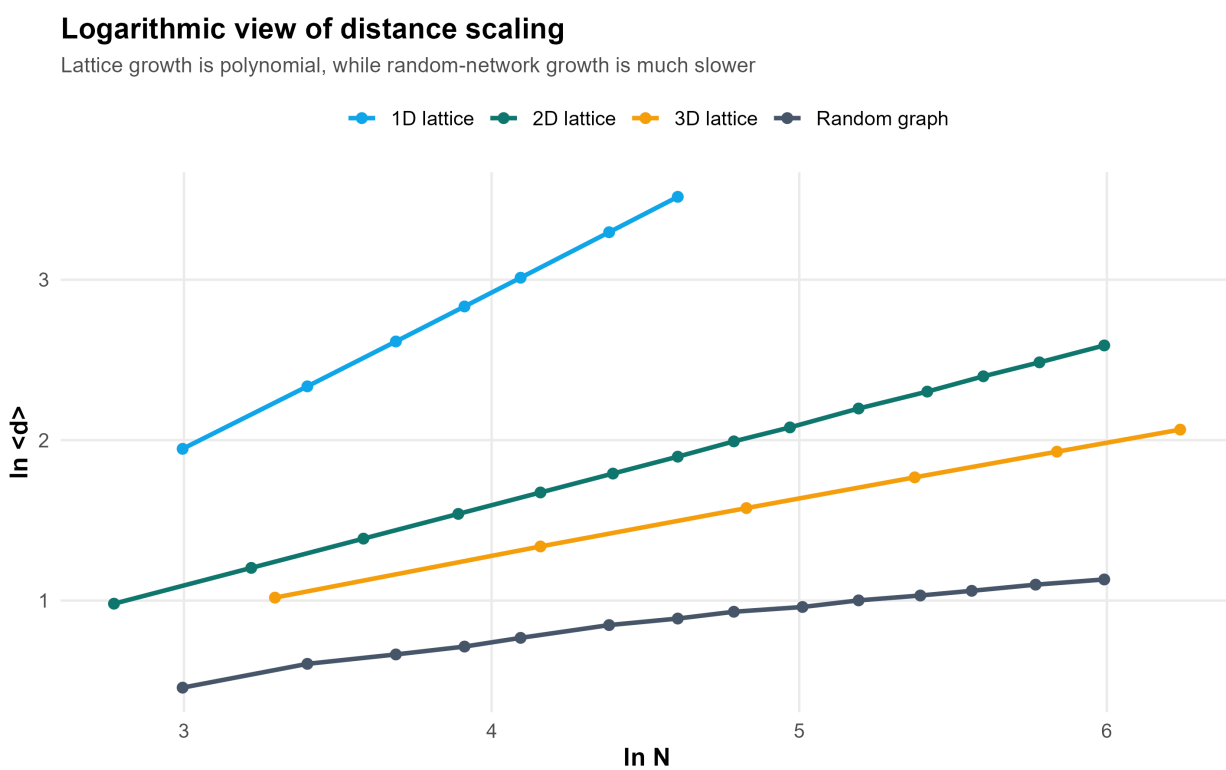


Figure 4.2: The same simulated data displayed as a plot of ln N versus ln .

These simulations reflect the expected asymptotic trends:

- the one-dimensional lattice grows approximately linearly in N,
- the two-dimensional lattice grows approximately like $N^{1/2}$,
- the three-dimensional lattice grows approximately like $N^{1/3}$,
- the random graph grows much more slowly and is consistent with logarithmic scaling.

This is the precise structural meaning of the small-world effect. In a random network, the typical separation between vertices increases so slowly with system size that even very large systems can remain metrically compact.

4.6 The Watts–Strogatz model

The random-network estimate above explains why short distances can occur, but it does not explain how a network can retain high clustering at the same time. That question is addressed by the Watts–Strogatz model.

4.6.1 Motivation

Two empirical observations often appear together in real networks:

1. the average distance between vertices is small and grows roughly logarithmically with system size,
2. the clustering coefficient is much higher than in a comparable random graph.

These two observations point in different structural directions.

- A **regular lattice** has strong local order and therefore high clustering, but it does not have short path lengths.
- A **random graph** has short path lengths, but its clustering is typically much smaller.

The Watts–Strogatz model was introduced to interpolate between these two extremes. Its purpose is to show that a graph can maintain strong local clustering while acquiring short global distances after only a small amount of random rewiring.

4.6.2 Parameters of the model

The standard parameters of the Watts–Strogatz model are:

- N: the number of vertices,
- K: the mean degree, assumed to be an even integer,
- $\beta \in [0, 1]$: the rewiring probability.

The parameter K must satisfy

$$2 \leq K < N, \qquad K \text{ even.}$$

In asymptotic discussions one often assumes

$$N \gg K \gg \ln N \gg 1,$$

because this is the regime in which the graph is sparse relative to the complete graph but still sufficiently connected to exhibit the small-world transition clearly.

4.6.3 Regular ring lattice

Definition 4.5. Let N and K be as above, with K even. The **regular ring lattice** is the graph on vertex set

$$V = \{0, 1, \dots, N-1\}$$

in which each vertex is connected to its $K/2$ nearest neighbors on each side around the ring.

Equivalently, vertices i and j are adjacent if and only if their cyclic distance on the ring is at most $K/2$.

Thus every vertex has degree exactly K, so the average degree of the lattice is

$$\langle k \rangle = K.$$

This graph is highly ordered. It has strong local structure and large clustering, but its average path length remains relatively large because travel across the graph requires many local steps.

4.6.4 Definition of the Watts–Strogatz model

In the original Watts–Strogatz construction, the rewiring probability is usually denoted by β.

Definition 4.6. The **Watts–Strogatz model** with parameters (N, K, β) is constructed as follows.

1. Start from the regular ring lattice on N vertices with degree K.
2. For each vertex
$$i = 0, 1, \dots, N-1,$$
consider the $K/2$ edges from i to its clockwise neighbors.
3. For each such edge, independently with probability β, rewire it by replacing its endpoint with a uniformly chosen new vertex.
4. During rewiring, self-loops and multiple edges are forbidden.

The parameter β is the **rewiring probability**.

A more explicit version of Step 2 is the following. For each vertex i, one considers the edges

$$(i, j) \qquad \text{such that} \qquad 0 < (j - i) \bmod N \leq \frac{K}{2}.$$

These are exactly the $K/2$ clockwise edges from i. Each one is then rewired independently with probability β.

4.6.5 Why only the clockwise edges are considered

It is important to notice that each undirected edge appears twice in the full ring description: once from each endpoint. To avoid treating the same undirected edge twice, the rewiring algorithm acts only on one oriented half of the local edges, namely the clockwise ones.

This makes the algorithm well defined while preserving the undirected nature of the final graph.

4.6.6 Interpretation of the parameter β

The parameter β controls the amount of randomness introduced into the graph.

- If
$$\beta = 0,$$
then no edge is rewired, so the graph remains the original regular lattice.
- If
$$0 < \beta \ll 1,$$
then only a small fraction of edges is rewired. Most local neighborhoods remain almost unchanged, but a few long-range shortcut edges appear.
- If
$$\beta = 1,$$
then every eligible edge has been rewired, so the graph becomes highly randomized.

Because rewiring preserves the total number of edges, the mean degree remains equal to K throughout the whole interpolation from order to randomness.

4.6.7 Ring-lattice benchmarks

For the regular ring lattice, one has

$$\langle k \rangle = K.$$

Its average path length grows approximately linearly with system size:

$$L(0) \approx \frac{N}{2K}.$$

Its clustering coefficient is also known explicitly:

$$C(0) = \frac{3(K-2)}{4(K-1)}.$$

This formula shows that the lattice has clustering of order one when K is fixed.

So the regular ring lattice has:

- large clustering,
- large average path length.

This is precisely why it is a useful starting point for the model.

```
library(igraph)
library(ggplot2)
library(dplyr)
library(purrr)
library(patchwork)
library(tibble)

make_ring_edges_K <- function(n, K) {
  stopifnot(K %% 2 == 0)
  tibble(
    from = rep(seq_len(n), each = K / 2),
    shift = rep(seq_len(K / 2), times = n)
  ) |>
    mutate(
      to = ((from + shift - 1) %% n) + 1
    ) |>
    select(from, to)
}

build_progressive_ws_graph_K <- function(n, K, beta, meta) {
  base_edges <- meta$base_edges
  adj <- matrix(FALSE, n, n)

  for (i in seq_len(nrow(base_edges))) {
    from <- base_edges$from[i]
    original_to <- base_edges$to[i]

    if (meta$u[i] <= beta) {
      candidates <- meta$candidates[[i]]
      idx <- which(!adj[from, candidates] & candidates != from)
      if (length(idx) == 0) {
```

```
        to <- original_to
      } else {
        to <- candidates[idx[1]]
      }
    } else {
      to <- original_to
    }

    adj[from, to] <- TRUE
    adj[to, from] <- TRUE
  }

  graph_from_adjacency_matrix(adj, mode = "undirected", diag = FALSE)
}
```

4.6.8 Visual interpolation from order to randomness

The next figure uses a fixed circular layout and a progressive rewiring construction so that the sequence looks like a genuine evolution from a regular lattice toward a random network.

Local lattice edges are shown in teal. Rewired long-range edges are shown in gold.

```
set.seed(2032)

n_vis <- 32
K_vis <- 4
beta_vis <- c(0, 0.01, 0.05, 0.20, 1.00)

base_edges_vis <- make_ring_edges_K(n_vis, K_vis)

ws_meta_vis <- list(
  base_edges = base_edges_vis,
  u = runif(nrow(base_edges_vis)),
  candidates = lapply(
    seq_len(nrow(base_edges_vis)),
    function(i) sample(setdiff(seq_len(n_vis), base_edges_vis$from[i]))
  )
)

# fixed circular layout
g0 <- build_progressive_ws_graph_K(n_vis, K_vis, 0, ws_meta_vis)
layout_ws <- layout_in_circle(g0)

nodes_df <- tibble(
  id = seq_len(n_vis),
  x = layout_ws[, 1],
  y = layout_ws[, 2]
)

make_edge_df <- function(g, K, nodes_df) {
  edge_pairs <- ends(g, E(g), names = FALSE) |>
    as.data.frame() |>
    setNames(c("from", "to")) |>
    as_tibble()
```

```r
  edge_pairs |>
    mutate(
      cyclic_dist = purrr::map2_dbl(from, to, \(a, b) {
        d <- abs(a - b)
        min(d, vcount(g) - d)
      }),
      edge_type = if_else(cyclic_dist <= K / 2, "local", "rewired")
    ) |>
    left_join(
      nodes_df |>
        rename(from = id, x = x, y = y),
      by = "from"
    ) |>
    left_join(
      nodes_df |>
        rename(to = id, xend = x, yend = y),
      by = "to"
    )
}

plot_progressive_ws_explicit <- function(beta_value, title_text) {
  g <- build_progressive_ws_graph_K(n_vis, K_vis, beta_value, ws_meta_vis)
  edges_df <- make_edge_df(g, K_vis, nodes_df)

  local_edges <- edges_df |> filter(edge_type == "local")
  rewired_edges <- edges_df |> filter(edge_type == "rewired")

  ggplot() +
    geom_curve(
      data = local_edges,
      aes(x = x, y = y, xend = xend, yend = yend),
      curvature = 0.38,
      linewidth = 0.55,
      colour = course_cols$teal,
      lineend = "round"
    ) +
    geom_curve(
      data = rewired_edges,
      aes(x = x, y = y, xend = xend, yend = yend),
      curvature = -0.18,
      linewidth = 0.8,
      colour = course_cols$gold,
      lineend = "round"
    ) +
    geom_point(
      data = nodes_df,
      aes(x = x, y = y),
      size = 2.8,
      shape = 21,
      fill = course_cols$blue,
      colour = "white",
      stroke = 0.5
    ) +
```

```r
    coord_equal(
      xlim = c(-1.08, 1.08),
      ylim = c(-1.08, 1.08),
      expand = FALSE,
      clip = "off"
    ) +
    labs(title = title_text) +
    theme_void() +
    theme(
      plot.title = element_text(
        hjust = 0.5,
        size = 10.5,
        face = "bold",
        margin = margin(b = 2)
      ),
      plot.margin = margin(0, 0, 0, 0)
    )
}

ws_plots <- map(
  beta_vis,
  ~ plot_progressive_ws_explicit(.x, paste0("\u03B2 = ", sprintf("%.2f", .x)))
)

wrap_plots(ws_plots, nrow = 1)
```

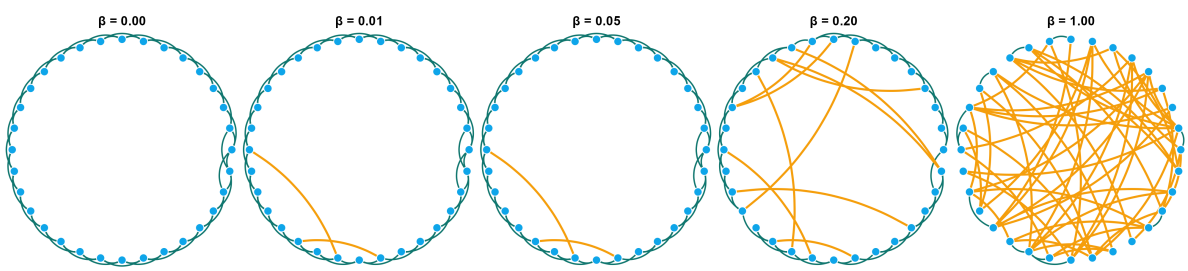


Figure 4.3: From regular lattice to random network in the Watts–Strogatz model. The same vertex positions are used throughout. Local edges are teal and rewired shortcut edges are gold.

This figure makes the mechanism of the model visually transparent.

- At $\beta = 0$, every edge is local and the graph is completely ordered.
- For small β, only a few edges are rewired, but they already connect distant parts of the ring.
- As β increases further, the number of shortcuts rises and the graph loses its regular geometric appearance.
- At $\beta = 1$, the graph is highly randomized.

The transition from order to randomness is therefore controlled continuously by the single parameter β.

4.6.9 Why shortcuts are so effective

A shortcut edge has a disproportionate effect on distance. In a ring lattice, moving from one side of the graph to the other requires many local steps. A single long-range edge can bypass a large segment of that travel.

So the reduction in path length is not proportional to the number of rewired edges. Even a very small rewiring probability can introduce enough shortcuts to lower distances dramatically.

By contrast, clustering is local. If only a small fraction of edges is rewired, then most neighborhoods still look almost like the original lattice, so clustering remains high.

This asymmetry is the key mathematical idea behind the small-world effect:

- **distance is highly sensitive to shortcuts**,
- **clustering is much less sensitive to a small amount of rewiring**.

! Core asymmetry of the Watts–Strogatz model

A small amount of random rewiring ($\beta \ll 1$) has a **disproportionate effect on path length** but leaves clustering largely intact.

- Path length drops sharply because a few long-range shortcuts bypass large ring segments.
- Clustering changes slowly because most local neighborhoods remain lattice-like.

This asymmetry is the structural source of the small-world regime.

4.6.10 Quantitative transition

To study the transition from order to randomness, one usually compares the average path length and clustering coefficient to their values at $\beta = 0$.

Define

$$\frac{L(\beta)}{L(0)}, \qquad \frac{C(\beta)}{C(0)}.$$

These normalized quantities are especially useful because they show how far the network has moved away from the regular lattice.

For a true small-world regime, one expects:

- a rapid decrease in $L(\beta)/L(0)$,
- while $C(\beta)/C(0)$ remains close to 1 over a substantially wider interval.

4.6.11 Numerical experiment

The next simulation follows a classical setup by using

$$N = 1000, \qquad K = 10.$$

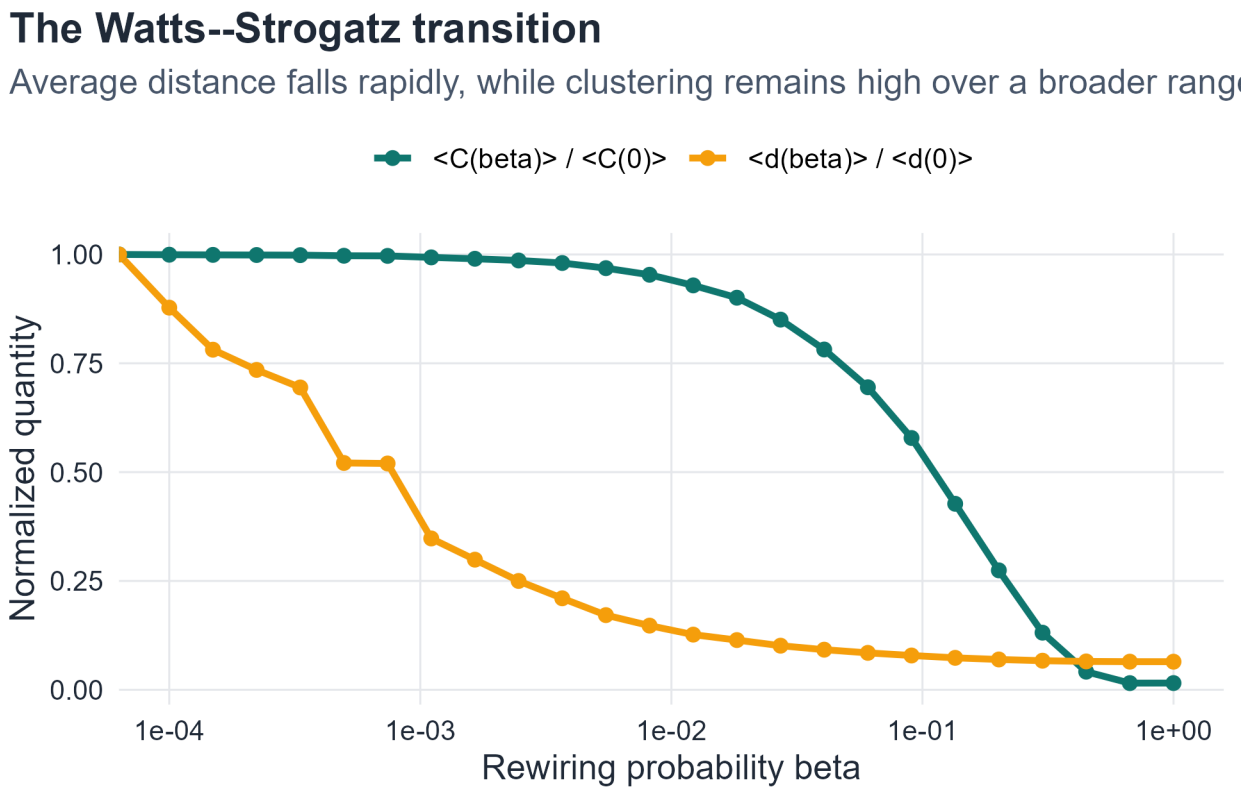


Figure 4.4: Normalized average path length and normalized clustering coefficient in the Watts–Strogatz model. The rapid drop in <d(beta)> marks the onset of the small-world regime.

This figure contains the main quantitative message of the model.

- The normalized average path length drops sharply even when β is still very small.
- The normalized clustering coefficient remains relatively high over a much wider range.

So there is an interval of rewiring probabilities for which the network has already acquired short path lengths, but still retains strong local clustering.

That interval is the **small-world regime**.

In many simulations of this type, that regime lies roughly between very small and moderate rewiring probabilities, often in a range comparable to

$$10^{-3} \lesssim \beta \lesssim 10^{-1},$$

although the precise interval depends on N and K.

4.7 Small-world regime

Definition 4.7. In the Watts–Strogatz model, the **small-world regime** is the range of rewiring probabilities β for which

1. the average path length is significantly smaller than in the regular lattice, and
2. the clustering coefficient remains substantially larger than in a comparable random graph.

This is not a single universal numerical threshold. It is a structural regime characterized by the simultaneous coexistence of:

- short global distances,
- strong local clustering.

That coexistence is what distinguishes a small-world network from both a regular lattice and a random graph.

Structural comparison

Model	Average path length	Clustering coefficient
Regular lattice	large ($\sim N/2K$)	high ($\sim 3/4$)
Random graph ($G(N,p)$)	small ($\sim \ln N/\ln\langle k\rangle$)	low ($\sim p$)
Watts–Strogatz (small-world)	small	high

Only the small-world model achieves both properties simultaneously.

4.8 What the Watts–Strogatz model explains and what it does not

The Watts–Strogatz model successfully explains one specific empirical coexistence:

- high clustering,
- short average path length.

This is a major achievement. But the model does **not** explain all structural properties observed in real networks.

In particular, its degree distribution remains relatively narrow. The rewiring mechanism preserves the number of edges and introduces only limited degree heterogeneity, so the model does not naturally generate hub-dominated, heavy-tailed degree distributions.

Therefore:

- the model captures the coexistence of clustering and short paths,
- but it does not explain the emergence of highly connected hubs.

Nor does it naturally reproduce the strong degree dependence of clustering often observed in hierarchical or scale-free empirical networks.

This limitation is conceptually important. The Watts–Strogatz model explains one aspect of real network structure, but not the full combination of features seen in many data sets.

4.9 Applications of small-world networks

The small-world concept is important not only as a mathematical idea, but also because many real systems appear to balance local organization with global reachability.

4.9.1 Social systems

In social networks, high clustering reflects the tendency of acquaintances of the same person to know one another. At the same time, short path lengths facilitate rapid information flow, rumor spreading, and social search.

So small-world structure helps explain how local communities can coexist with short global separation.

4.9.2 Neural systems

In neural and brain networks, local clustering is associated with local specialization and dense intra-regional processing, while short path lengths support efficient long-range communication across distant regions.

This combination is often interpreted as a compromise between segregation and integration.

4.9.3 Transportation and infrastructure

Airline networks, road systems with hubs, and infrastructure networks often require efficient travel between distant points while maintaining strong local substructure. Shortcut-like edges can reduce travel time significantly without eliminating regional organization.

4.9.4 Communication and information systems

In communication networks and technological networks, short path lengths support efficient routing, synchronization, and dissemination. At the same time, local clustering may reflect modular architecture or design constraints.

4.9.5 Why applications matter

The practical importance of small-world structure is rooted in a trade-off:

- purely local systems are well organized but globally slow,
- purely random systems are globally efficient but may lack local coherence,
- small-world systems can combine the advantages of both.

This is why the Watts–Strogatz model remains one of the most influential models in network science.

4.10 Core formulas to remember

! Summary

For small-world networks and the Watts–Strogatz model:

1. **Average path length (random network):**

$$L(G) \approx \frac{\ln N}{\ln\langle k\rangle}$$

2. **Lattice distance scaling** (d-dimensional):

$$L(G) \sim D(G) \sim N^{1/d}$$

3. **Ring-lattice average path length:**

$$L(0) \approx \frac{N}{2K}$$

4. **Ring-lattice clustering coefficient:**

$$C(0) = \frac{3(K-2)}{4(K-1)}$$

5. **Small-world regime** (β small but nonzero):

$$\frac{L(\beta)}{L(0)} \ll 1, \qquad \frac{C(\beta)}{C(0)} \approx 1$$

6. **Expected clustering in** $G(N,p)$:

$$\mathbb{E}[C] = p \ll C(0)$$

4.11 In-class questions

1. What does "small" mean mathematically in the phrase *small-world*?
2. Why is the estimate $\ln N / \ln\langle k\rangle$ associated with random networks?
3. Why is the small-world effect surprising from the viewpoint of lattice geometry?
4. Why do shortcuts affect path length more strongly than clustering?
5. Why is the Watts–Strogatz model structurally different from an Erdős–Rényi random graph?
6. Why is average path length usually more stable than diameter?
7. What does the Watts–Strogatz model explain successfully, and what does it fail to explain?
8. Why are small-world structures useful in applications?

4.12 End-of-chapter exercises

The exercises below are divided into three groups:

1. mathematical exercises,
2. conceptual and analytical exercises,
3. simulation and coding exercises.

4.12.1 Part I. Mathematical exercises

4.12.1.1 Exercise 1. Geometric-series derivation

Starting from the branching approximation

$$N(d) \approx 1 + \langle k\rangle + \langle k\rangle^2 + \cdots + \langle k\rangle^d,$$

derive the closed-form expression

$$N(d) \approx \frac{\langle k\rangle^{d+1} - 1}{\langle k\rangle - 1}.$$

State clearly the assumption under which this approximation is meaningful.

4.12.1.2 Exercise 2. Logarithmic distance estimate

Starting from

$$N(d_{\max}) \approx N,$$

derive the estimate

$$d_{\max} \approx \frac{\ln N}{\ln\langle k\rangle}.$$

Explain each algebraic step carefully.

4.12.1.3 Exercise 3. Why the estimate is not exact

The relation

$$L(G) \approx \frac{\ln N}{\ln\langle k\rangle}$$

is only an approximation.

Explain why it is not exact. Your answer should discuss:

- overlap between neighborhoods,
- loops and finite-size effects,
- the difference between average distance and diameter.

4.12.1.4 Exercise 4. Lattice scaling laws

Show that in a d-dimensional regular lattice with N vertices, the characteristic linear size of the system is of order

$$N^{1/d}.$$

Explain why this leads to the scaling

$$L(G) \sim D(G) \sim N^{1/d}.$$

4.12.1.5 Exercise 5. Average degree in the ring lattice

Let G be the regular ring lattice with parameters (N, K).

1. Prove that every vertex has degree K.
2. Deduce that
$$|E| = \frac{NK}{2}.$$
3. Compute the average degree.

4.12.1.6 Exercise 6. Normalized observables

Let $C(\beta)$ and $L(\beta)$ be the clustering coefficient and average path length in the Watts–Strogatz model.

1. Define the normalized quantities
$$\frac{C(\beta)}{C(0)}, \qquad \frac{L(\beta)}{L(0)}.$$
2. Explain why these ratios are useful when studying the transition from order to randomness.

3. State the behavior expected in the small-world regime.

4.12.1.7 Exercise 7. Ring-lattice path length

Give a heuristic derivation of the estimate

$$L_{\text{lattice}} \approx \frac{N}{2K}$$

for the regular ring lattice.

4.12.1.8 Exercise 8. Ring-lattice clustering

Using the ring lattice with parameter K, derive the formula

$$C(0) = \frac{3(K-2)}{4(K-1)}.$$

4.12.2 Part II. Conceptual and analytical exercises

4.12.2.1 Exercise 9. Six degrees and logarithmic scaling

Explain why the statement "all vertices are separated by only a few steps" should not be interpreted as a fixed constant independent of system size.

Your answer should explain the difference between: - a constant bound, - logarithmic growth.

4.12.2.2 Exercise 10. Why small worlds are surprising

Write a short mathematical discussion comparing the growth laws

$$\ln N, \qquad N^{1/3}, \qquad N^{1/2}, \qquad N.$$

Explain why the logarithmic law is fundamentally different from polynomial lattice scaling.

4.12.2.3 Exercise 11. Random graph versus small-world network

A random graph and a Watts–Strogatz network can both have short average path length.

Explain why they should still be considered structurally different models.

Your answer should address: - clustering, - local order, - global reachability.

4.12.2.4 Exercise 12. Why shortcuts are powerful

Explain why a small number of long-range edges can drastically reduce average path length.

Your answer should use the language of graph distance and should distinguish between local and long-range edges.

4.12.2.5 Exercise 13. Limitations of the Watts–Strogatz model

Discuss why the Watts–Strogatz model does not naturally explain hub-dominated degree distributions or strong degree-dependent clustering.

4.12.2.6 Exercise 14. Applications of small-world structure

Choose two application domains from the chapter and explain why the coexistence of high clustering and short path lengths is useful in each one.

4.12.3 Part III. Simulation and coding exercises

4.12.3.1 Exercise 15. Distance scaling in random networks

For several values of N and fixed average degree, simulate random graphs and estimate the average path length.

Compare the empirical results with the prediction

$$L(G) \approx \frac{\ln N}{\ln \langle k \rangle}.$$

Discuss the quality of the approximation.

4.12.3.2 Exercise 16. Simulated scaling across network families

Reproduce the comparison of average path length for:

- 1D lattices,
- 2D lattices,
- 3D lattices,
- random graphs.

Display the results both:

- on ordinary axes,
- on logarithmic axes.

Then interpret the observed scaling.

4.12.3.3 Exercise 17. Small-world transition plot

For fixed N and K, simulate the Watts–Strogatz model over a logarithmic grid of β values.

1. compute $C(\beta)$ and $L(\beta)$,
2. normalize both by their values at $\beta = 0$,
3. plot the normalized quantities on the same axes,
4. identify the approximate small-world regime.

4.12.3.4 Exercise 18. Lattice versus random versus small-world

Generate three graphs with comparable average degree:

1. a ring lattice,
2. a Watts–Strogatz graph with small rewiring probability,
3. an Erdős–Rényi random graph.

For each graph, compute: - average degree, - clustering coefficient, - average path length.

Compare the results and discuss the structural differences.

4.12.3.5 Exercise 19. Fixed-layout visualization

Construct a fixed-layout sequence of Watts–Strogatz graphs for increasing values of β.

1. Use identical vertex positions in all figures.
2. Highlight shortcut edges differently from local edges.
3. Describe how the graph changes visually.
4. Explain where the small-world effect becomes visible.

4.12.3.6 Exercise 20. Finite-size comparison

Repeat the normalized-clustering and normalized-path-length simulation for several values of N.

Discuss how the location and width of the small-world regime change, or remain stable, as the system size increases.

4.12.4 Suggested mini-project

A strong mini-project for this chapter is the following:

> Compare the distance structure of regular lattices, random graphs, and Watts–Strogatz networks under matched average degree.

A complete submission should contain:

- mathematical derivations,
- simulation code,
- at least two figures,
- one discussion of logarithmic versus polynomial distance scaling,
- one discussion of the Watts–Strogatz transition,
- a concluding discussion of the small-world effect.

4.13 Concluding remarks

The small-world phenomenon means that the typical distance between vertices in a large network can be unexpectedly small. In random networks this behavior is explained by logarithmic distance scaling. In regular lattices it is absent, because distances grow polynomially with system size.

The Watts–Strogatz model shows how these two worlds can be connected. A small amount of random rewiring introduces shortcuts that rapidly reduce path lengths while leaving most local clustering intact.

The central lesson of the chapter is therefore this: short global separation and strong local order can coexist in the same network, and the small-world model provides a concrete mechanism for that coexistence.

Chapter 5

Scale-Free Networks

5.1 Introduction

The Erdős–Rényi model and the Watts–Strogatz model explain important aspects of network structure, but many real systems display a feature that neither model captures naturally: the existence of a small number of highly connected vertices, usually called **hubs**.

Examples include:

- highly cited scientific papers,
- very popular web pages,
- airports with exceptionally high traffic,
- social actors with unusually many connections.

In such systems, degree is not concentrated around a typical value. Instead, the degree distribution is strongly heterogeneous. Most vertices have small degree, but a few vertices have extremely large degree. This broad distribution is one of the defining signatures of a scale-free network.

The purpose of this chapter is to study the mathematical meaning of scale-free behavior and the mechanism by which it can arise. The central model is the **Barabási–Albert model**, which combines two key ingredients:

1. growth,
2. preferential attachment.

Together they generate networks with heavy-tailed degree distributions and hub formation.

5.2 Learning goals

By the end of this chapter, it should be possible to:

- define degree heterogeneity and hub structure
- define a power-law degree distribution
- explain what is meant by a scale-free network
- define preferential attachment mathematically
- describe the Barabási–Albert model
- explain why preferential attachment leads to hub formation
- verify scale-free behavior numerically in `R`

5.3 Broad degree distributions and hubs

5.3.1 Degree heterogeneity

A first step toward scale-free structure is to understand that degree distributions need not be concentrated around one typical value.

In a homogeneous network, most vertices have degree values of roughly the same order, and the distribution is relatively narrow. By contrast, in a heterogeneous network, degree values span many scales.

Definition 5.1. A network is said to exhibit **degree heterogeneity** if its degree distribution is broad, in the sense that vertices of very different degree coexist in significant numbers.

This definition is qualitative, but it captures an essential structural distinction. In a heterogeneous network, one cannot meaningfully describe the graph using only a "typical" degree.

5.3.2 Hubs

Definition 5.2. A **hub** is a vertex whose degree is substantially larger than the degrees of most other vertices in the same network.

The notion of a hub is relative to the graph under consideration. A vertex of degree 20 may be a hub in one network and completely ordinary in another.

The significance of hubs is not merely visual. They strongly influence:

- shortest paths,
- robustness,
- spreading processes,
- centralization of connectivity.

So the appearance of hubs is one of the main reasons why scale-free networks differ from more homogeneous random graphs.

5.3.3 Visual contrast: homogeneous and heterogeneous degree structure

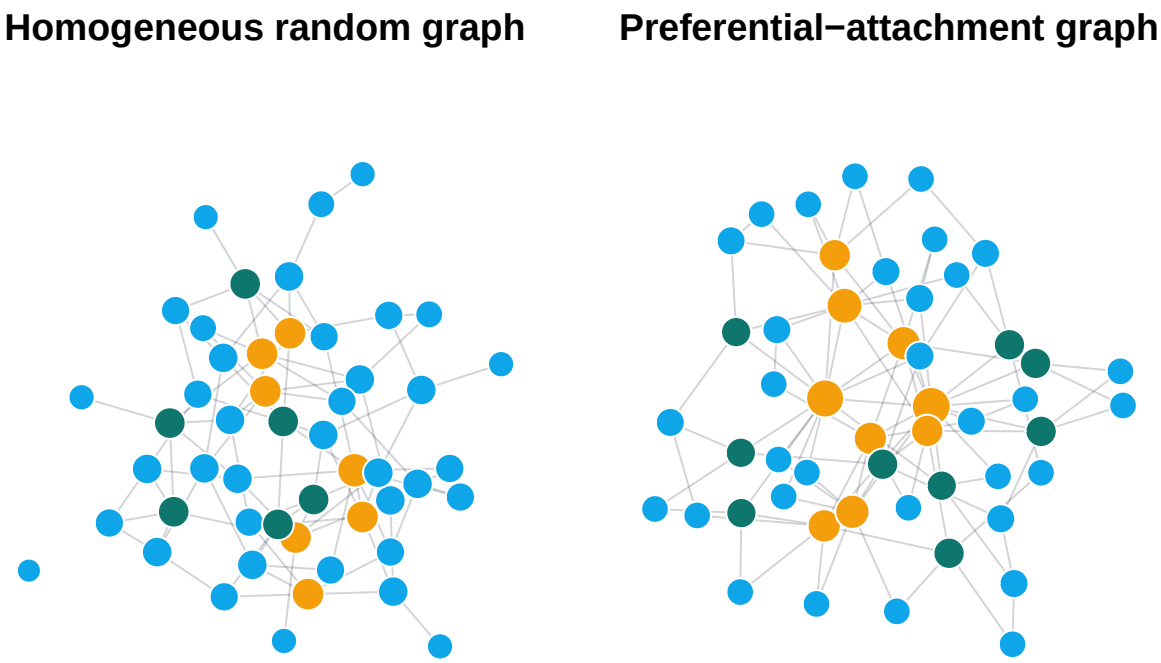


Figure 5.1: A comparison between a homogeneous random graph and a heterogeneous preferential-attachment graph with approximately similar average degree.

The comparison above illustrates the basic distinction:

- in a homogeneous random graph, vertex degrees tend to lie in a relatively narrow range
- in a preferential-attachment graph, a few vertices accumulate much larger degree and emerge as hubs

This does not yet prove a power law, but it motivates the need for a more precise mathematical description.

5.4 Power-law degree distributions

5.4.1 Definition of a power law

Definition 5.3. A degree distribution is said to follow a **power law** if, for sufficiently large degree k,

$$P(k) \sim Ck^{-\gamma},$$

where $C > 0$ is a constant and $\gamma > 0$ is the degree exponent.

The notation $\sim$ means asymptotic proportionality. More precisely,

$$\lim_{k\to\infty} \frac{P(k)}{Ck^{-\gamma}} = 1.$$

So a power law describes the tail behavior of the degree distribution.

5.4.2 Why the term "scale-free" is used

Suppose

$$P(k) \sim Ck^{-\gamma}.$$

Then for any scaling factor $a > 0$,

$$P(ak) \sim C(ak)^{-\gamma} = a^{-\gamma}Ck^{-\gamma}.$$

So changing the scale of k only multiplies the distribution by a constant factor. There is no preferred characteristic scale in the tail. This is the reason for the term **scale-free**.

Definition 5.4. A network is called **scale-free** if its degree distribution has a power-law tail over a substantial range of degrees.

This definition should be interpreted with care. In applications, one usually does not expect a perfect power law over all degrees. What matters is that the tail decays approximately as a power law over a meaningful interval.

5.4.3 Cumulative distribution

In empirical work it is often useful to consider the cumulative degree distribution

$$P(K \geq k).$$

If

$$P(k) \sim Ck^{-\gamma},$$

then the cumulative distribution satisfies

$$P(K \geq k) \sim C'k^{-(\gamma-1)}$$

for large k, provided $\gamma > 1$.

This representation is often smoother numerically and easier to visualize on a log-log scale.

5.5 Preferential attachment

The central mechanism behind the Barabási–Albert model is preferential attachment.

Definition 5.5. A growing network is said to evolve by **preferential attachment** if the probability that a new vertex connects to an existing vertex i is proportional to the degree of i.

If the degree of vertex i at time t is denoted by $k_i(t)$, then preferential attachment takes the form

$$\Pi_i(t) = \frac{k_i(t)}{\sum_j k_j(t)}.$$

This means that highly connected vertices are more likely to receive new links. In informal language, "the rich get richer."

5.5.1 Interpretation

Preferential attachment is not merely a mathematical trick. It encodes a plausible dynamic for many real systems.

- a popular web page is more likely to attract new links,
- a highly cited article is more likely to receive future citations,
- a well-connected airport is more likely to gain new routes,
- a socially visible person is more likely to form further connections.

So preferential attachment captures a cumulative-advantage mechanism.

> **Key result: preferential attachment rule**
>
> In a growing network with preferential attachment, the probability that a new vertex connects to existing vertex i is
>
> $$\Pi_i(t) = \frac{k_i(t)}{\sum_j k_j(t)}.$$
>
> Since $\sum_j k_j(t) = 2L(t)$, this simplifies to $\Pi_i(t) = k_i(t)/2L(t)$. A vertex with twice the degree is twice as likely to receive a new edge.

5.5.2 Why preferential attachment matters

If edges were added uniformly at random to existing vertices, the graph would not generate strong hub structure in the same way. Preferential attachment amplifies pre-existing degree differences.

Once a vertex gains an advantage, it becomes more likely to gain future links, which can strengthen the advantage further. This feedback is the source of hub formation.

5.6 The Barabási–Albert model

Definition 5.6. The **Barabási–Albert model** is defined by the following growth rule:

1. begin from a connected seed graph,
2. at each time step add one new vertex,
3. connect the new vertex to m existing vertices,
4. choose each target vertex with probability proportional to its degree.

The model therefore combines two ingredients:

- **growth**: the number of vertices increases over time
- **preferential attachment**: attachment probability is degree-proportional

Both ingredients are essential. Growth alone does not produce scale-free structure, and preferential choice without growth does not generate the same degree dynamics.

5.6.1 Average degree in the BA model

At each time step, one new vertex and m new edges are added. For large network size, the average degree tends to

$$\langle k \rangle \approx 2m.$$

This follows from the fact that each new edge contributes 2 to the total degree and that the number of edges grows linearly with the number of added vertices.

So the parameter m controls the density of the graph.

5.7 Growth of a preferential-attachment network

The next figure illustrates the evolution of a Barabási–Albert network as new vertices are added.

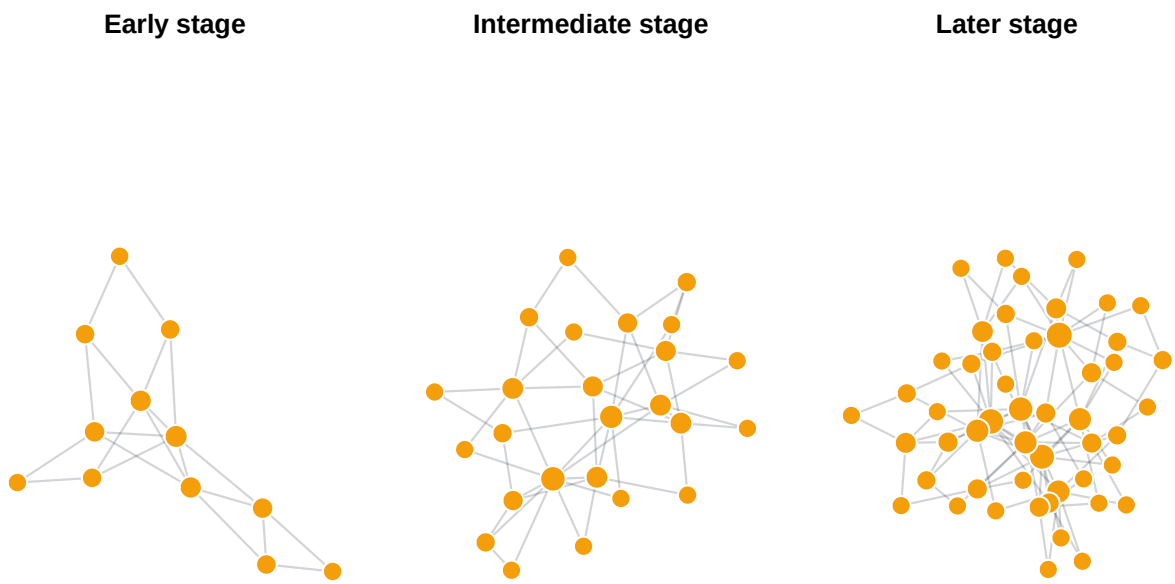


Figure 5.2: Growth of a preferential-attachment network. Older highly connected vertices begin to dominate as the network expands.

This figure shows the central mechanism of the model:

- early fluctuations matter,
- some vertices acquire an advantage early,
- that advantage is amplified over time,
- hubs gradually emerge.

So the model does not treat all vertices symmetrically across time. Age and accumulated degree both influence network structure.

5.8 Degree distribution in the BA model

A classical result for the Barabási–Albert model is that the degree distribution follows a power law with exponent

$$\gamma = 3.$$

More precisely, in the asymptotic limit,

$$P(k) \sim Ck^{-3}.$$

This means that the degree distribution is heavy-tailed: high-degree vertices are rare, but much more common than in models with exponentially decaying tails.

5.8.1 Consequences of a heavy tail

The power-law form has several important structural implications.

1. Most vertices have small degree.
2. A few vertices may have very large degree.
3. There is no narrow characteristic degree scale in the tail.
4. Hub structure becomes a natural outcome of the dynamics.

This is what makes scale-free networks qualitatively different from Erdős–Rényi random graphs, whose degree distributions are much more concentrated.

! Classical result: BA degree distribution

The Barabási–Albert model with parameter m produces a power-law degree distribution with exponent $\gamma = 3$:

$$P(k) \sim Ck^{-3}, \qquad k \gg m.$$

This result is independent of m. The parameter m controls the density ($\langle k \rangle \approx 2m$) but not the tail exponent. Heavy tails are an inevitable consequence of the growth-plus-preferential-attachment mechanism.

5.9 Numerical exploration

5.9.1 Degree distribution of a BA network

```r
set.seed(2040)

n_ba <- 3000
m_ba <- 2

g_ba_large <- sample_pa(n = n_ba, power = 1, m = m_ba, directed = FALSE)
deg_ba <- degree(g_ba_large)

deg_tbl <- tibble(k = deg_ba) |>
  count(k, name = "count") |>
  filter(k > 0) |>
  mutate(
    prob = count / sum(count)
  )

deg_tbl
```

```
# A tibble: 37 x 3
       k count    prob
   <dbl> <int>   <dbl>
 1     2  1361 0.454
 2     3   635 0.212
 3     4   331 0.110
 4     5   202 0.0673
 5     6   150 0.05
 6     7    66 0.022
 7     8    52 0.0173
 8     9    32 0.0107
 9    10    29 0.00967
10    11    29 0.00967
# i 27 more rows
```

```r
ggplot(deg_tbl, aes(x = k, y = prob)) +
  geom_point(color = course_cols$teal, size = 2.0, alpha = 0.8) +
  scale_x_log10() +
  scale_y_log10() +
  labs(
    title = "Degree distribution of a preferential-attachment network",
    subtitle = "The heavy tail is visible on log-log axes",
    x = "Degree k",
    y = "P(k)"
  )
```

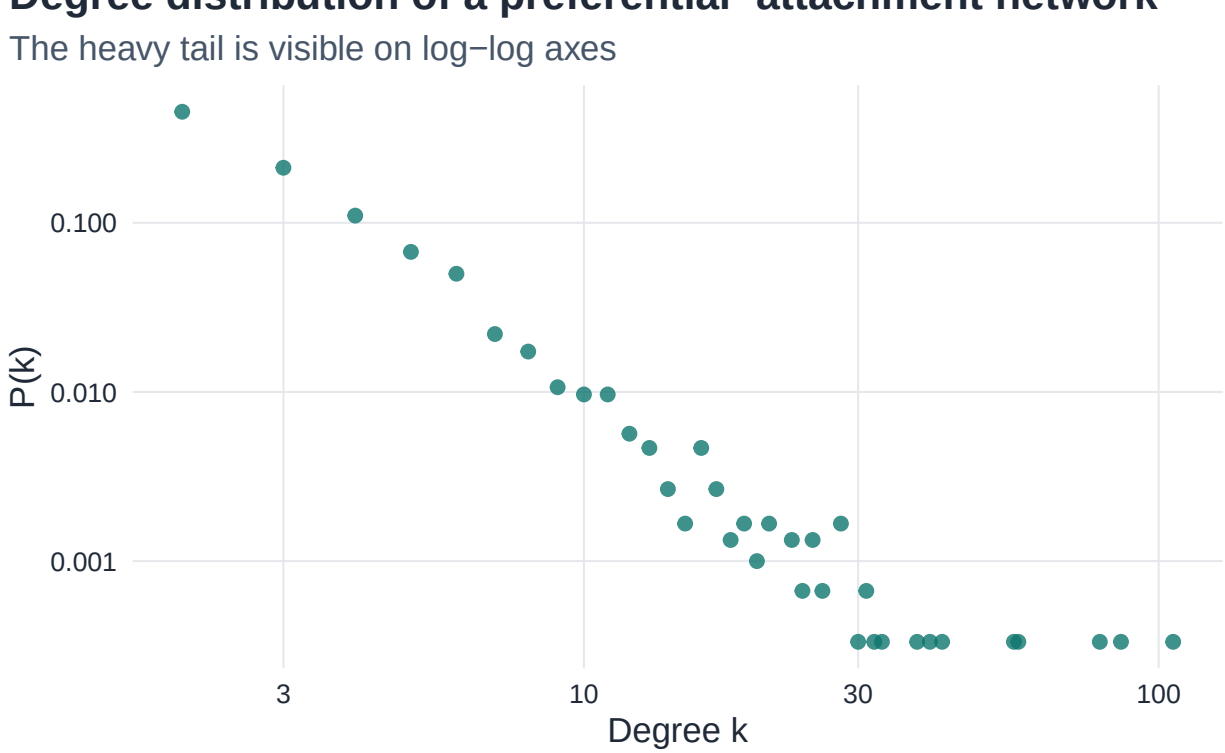


Figure 5.3: Empirical degree distribution of a Barabási–Albert network on log-log axes.

This plot does not by itself prove a power law, but it shows the heavy-tailed structure clearly. The tail decays much more slowly than the binomial or Poisson degree distributions seen in homogeneous random graphs.

5.9.2 Cumulative degree distribution

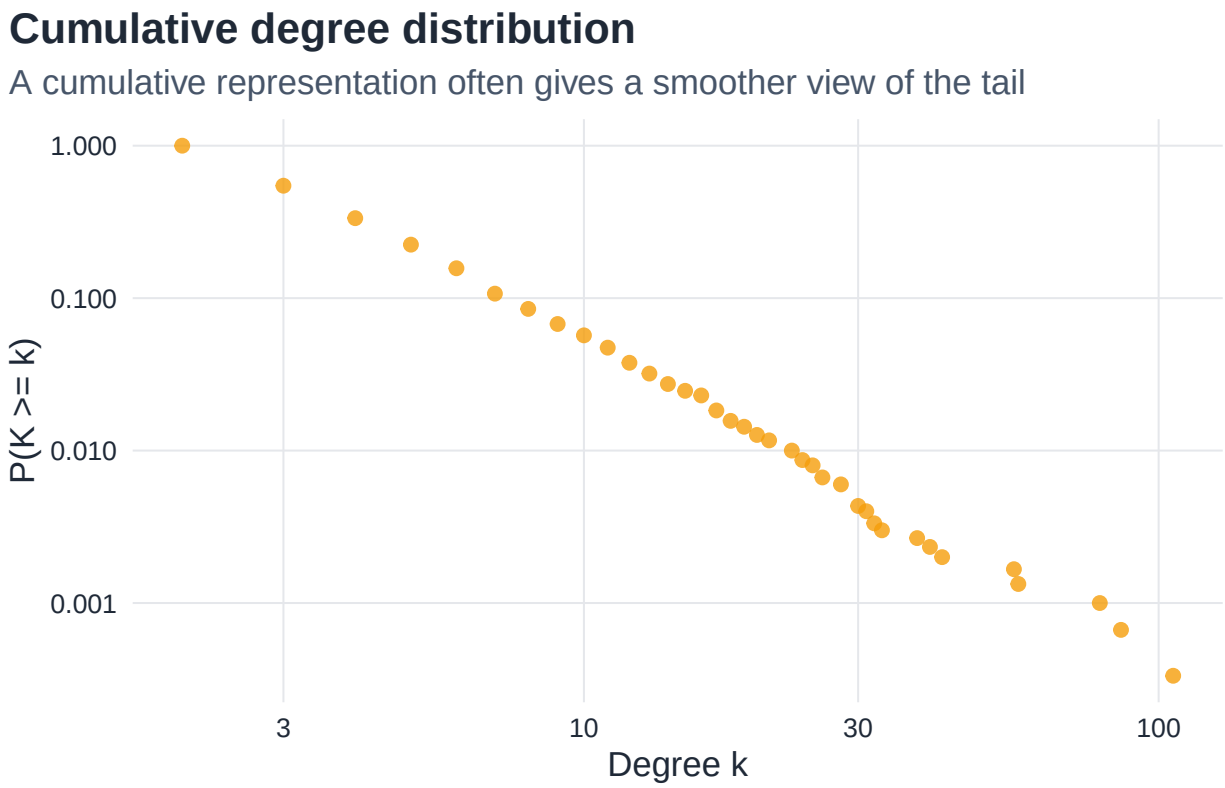


Figure 5.4: Empirical cumulative degree distribution of a Barabási–Albert network on log-log axes.

The cumulative distribution is often smoother than the raw degree histogram and is therefore useful when investigating heavy-tailed behavior.

5.10 Comparison with Erdős–Rényi graphs

A direct comparison with an Erdős–Rényi graph having approximately the same average degree helps clarify the difference in degree structure.

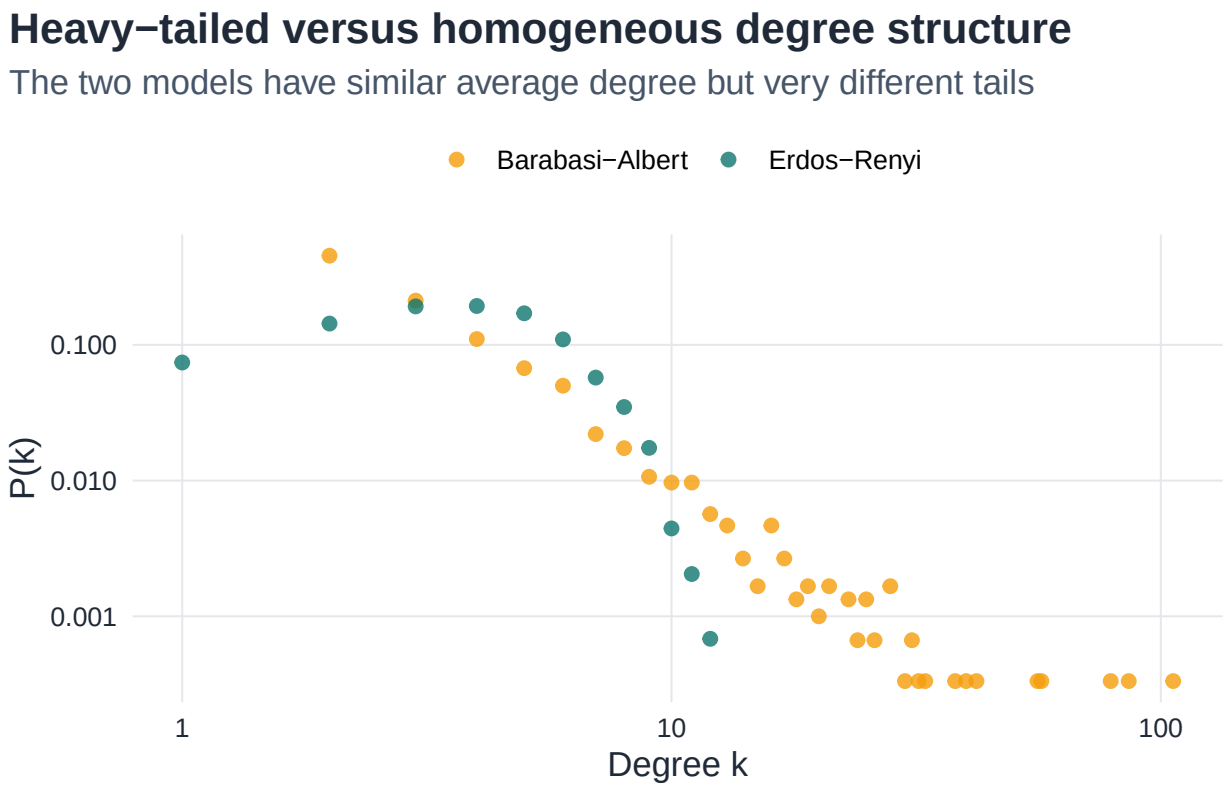


Figure 5.5: Degree distributions of a Barabási–Albert graph and an Erdős–Rényi graph with approximately matched average degree.

The contrast is the essential one:

- the Erdős–Rényi graph has a narrow degree distribution,
- the Barabási–Albert graph has a broad heavy-tailed degree distribution.

This is the mathematical signature of hub formation.

5.11 Why scale-free structure is important

Scale-free structure matters because degree heterogeneity strongly affects network behavior.

5.11.1 Paths and centralization

Hubs can act as highly efficient bridges. They reduce many shortest paths and create strong centralization of flow.

5.11.2 Robustness and vulnerability

A network with hubs may be tolerant to random deletion of vertices, because most randomly chosen vertices are low-degree. But it may be highly vulnerable to targeted removal of hubs.

5.11.3 Spreading processes

Infections, information, or cascades can spread differently in hub-dominated networks than in homogeneous graphs, because hubs can act as powerful transmission centers.

So the shape of the degree distribution is not merely descriptive. It has dynamical consequences.

5.12 Core formulas to remember

! Summary

For scale-free networks and the Barabási–Albert model:

1. **Preferential attachment probability:**

$$\Pi_i(t) = \frac{k_i(t)}{\sum_j k_j(t)} = \frac{k_i(t)}{2L(t)}$$

2. **Average degree in the BA model:**

$$\langle k \rangle \approx 2m$$

3. **Power-law degree distribution:**

$$P(k) \sim Ck^{-\gamma}, \qquad \gamma = 3 \text{ for the BA model}$$

4. **Scale-free property:**

$$P(ak) = a^{-\gamma} P(k) \quad \text{for all } a > 0$$

5. **Cumulative tail:**

$$P(K \geq k) \sim C' k^{-(\gamma-1)}$$

6. **Degree distribution comparison:**

Model	Degree distribution	Typical form
$G(N, p)$	Binomial / Poisson	narrow, peaked
Watts–Strogatz	Near-Poisson	narrow, peaked
Barabási–Albert	Power law	heavy-tailed, hub-dominated

5.13 In-class questions

1. Why does a broad degree distribution imply that average degree alone is insufficient to describe the graph?
2. What is the difference between a heavy-tailed degree distribution and a narrow one?
3. Why does preferential attachment lead naturally to hub formation?
4. Why are growth and preferential attachment both essential in the BA model?
5. Why does a scale-free network differ structurally from an Erdős–Rényi graph even when their average degrees are similar?

5.14 End-of-chapter exercises

The exercises below are divided into three groups:

1. mathematical exercises,
2. conceptual and analytical exercises,
3. simulation and coding exercises.

5.14.1 Part I. Mathematical exercises

5.14.1.1 Exercise 1. Average degree in the BA model

In the Barabási–Albert model, one new vertex and m new edges are added at each time step.

1. Let N be the number of vertices after many growth steps. Show that the total number of edges is approximately proportional to mN.
2. Deduce that the average degree is approximately
$$\langle k \rangle \approx 2m.$$
3. Explain why this approximation becomes accurate for large N.

5.14.1.2 Exercise 2. Preferential-attachment probability

Suppose the degree of vertex i at time t is $k_i(t)$.

1. Explain why the preferential-attachment rule can be written as
$$\Pi_i(t) = \frac{k_i(t)}{\sum_j k_j(t)}.$$
2. Show that
$$\sum_i \Pi_i(t) = 1.$$
3. Interpret this expression probabilistically.

5.14.1.3 Exercise 3. Power-law scaling

Assume that the degree distribution satisfies

$$P(k) \sim Ck^{-\gamma}.$$

1. Show that
$$P(ak) \sim a^{-\gamma} P(k)$$
for every fixed $a > 0$.
2. Explain why this motivates the term *scale-free*.
3. Discuss why such a distribution does not possess a single characteristic tail scale.

5.14.1.4 Exercise 4. Cumulative power-law tail

Assume that

$$P(k) \sim Ck^{-\gamma}$$

with $\gamma > 1$.

Show heuristically that the cumulative tail satisfies

$$P(K \geq k) \sim C' k^{-(\gamma-1)}.$$

Explain why cumulative distributions are useful in empirical analysis.

5.14.1.5 Exercise 5. Hub interpretation

Let $k_{\max}$ denote the largest observed degree in a graph.

1. Explain why a graph with narrow degree distribution typically has moderate $k_{\max}$.
2. Explain why a heavy-tailed degree distribution allows much larger hub degrees.
3. Discuss why hub formation is a consequence of broad degree heterogeneity.

5.14.1.6 Exercise 6. BA exponent

A classical result for the Barabási–Albert model is

$$P(k) \sim Ck^{-3}.$$

Explain the meaning of this statement and discuss how it differs conceptually from the binomial or Poisson degree laws encountered in earlier chapters.

5.14.2 Part II. Conceptual and analytical exercises

5.14.2.1 Exercise 7. Growth versus static models

Write a short analytical comparison between:

- the Erdős–Rényi model,
- the Watts–Strogatz model,
- the Barabási–Albert model.

Your answer should address: - whether the model is static or growing, - whether degree distribution is narrow or broad, - whether clustering is naturally high, - whether hubs appear naturally.

5.14.2.2 Exercise 8. Why preferential attachment creates hubs

Give a careful explanation of why preferential attachment amplifies degree differences over time.

Your answer should address: - cumulative advantage, - positive feedback, - the role of early vertices.

5.14.2.3 Exercise 9. Scale-free versus small-world

Explain why the terms *scale-free* and *small-world* describe different properties.

Your answer should explain: - what structural quantity each concept refers to, - why a network can be scale-free without being a Watts–Strogatz network, - why a network can be small-world without having a power-law degree distribution.

5.14.2.4 Exercise 10. Why average degree is insufficient

Two graphs may have the same average degree but very different structure.

Discuss this statement in the context of: - Erdős–Rényi graphs, - Barabási–Albert graphs.

Your answer should explain why average degree alone does not capture heterogeneity or hub structure.

5.14.3 Part III. Simulation and coding exercises

5.14.3.1 Exercise 11. Simulating the BA model

Write an `R` function that generates a Barabási–Albert graph with parameters (N, m) and returns:

- the average degree,
- the maximum degree,
- the degree sequence,
- the empirical degree distribution.

Test the function for several values of N and m.

5.14.3.2 Exercise 12. Log-log degree plot

For a fixed Barabási–Albert graph:

1. compute the empirical degree distribution,
2. display it on ordinary axes,
3. display it on log-log axes,
4. explain why the heavy tail is more visible on log-log coordinates.

5.14.3.3 Exercise 13. Cumulative distribution

For a Barabási–Albert graph with large N:

1. compute the empirical cumulative degree distribution,
2. plot it on log-log axes,
3. compare it with the raw empirical degree distribution,
4. discuss which representation is smoother and why.

5.14.3.4 Exercise 14. Comparison with Erdős–Rényi

Generate:

- one Barabási–Albert graph with parameters (N, m),
- one Erdős–Rényi graph with approximately the same average degree.

Then compare:

- average degree,
- maximum degree,
- degree distribution,
- visual appearance.

Discuss why the two networks differ despite matched average degree.

5.14.3.5 Exercise 15. Effect of system size

Repeat the BA simulation for several values of N.

Study how the following quantities change with N:

- maximum degree,
- degree heterogeneity,

- visibility of the heavy tail.

Explain why hub structure becomes more pronounced as the network grows.

5.14.4 Suggested mini-project

A strong mini-project for this chapter is the following:

> Compare the degree structure of homogeneous random graphs and preferential-attachment networks under matched average degree.

A complete submission should contain:

- mathematical definitions,
- simulation code,
- at least two quantitative figures,
- a comparison of maximum degree and degree distribution,
- a concluding discussion of hub formation.

5.15 Concluding remarks

Scale-free networks are characterized by strong degree heterogeneity and the emergence of hubs. Their degree distributions are fundamentally different from the narrow distributions of homogeneous random graphs.

The Barabási–Albert model explains one mechanism capable of producing such structure: growth combined with preferential attachment. This mechanism shows how large inequalities in connectivity can emerge naturally from a simple probabilistic rule.

The central lesson of the chapter is that network evolution rules shape degree structure, and that broad degree distributions lead to qualitatively different network organization.